



## MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

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# DETAILED SYLLABUS

## First Year — Semesters I

*Common to all Bachelor of Technology programmes of the University*

Prepared against the finalised NEP 2020 Compliant Curriculum Structure for the First Year  
**(B.Tech. First Year, MAKAUT 2026 — Effective from Academic Year 2026–27)**

Outcome-Based · NEP 2020 · NAAC and NBA Aligned · Eleven Programme Outcomes (NBA GAPC v4.0)

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## Contents

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**Foreword** — *What this document is, and the changes carried into this redesign*

**Part I – The Outcome-Based Framework** — *The eleven Programme Outcomes (NBA GAPC v4.0), Bloom levels, articulation, attainment*

**Part II – The Bridge from Class XII** — *Prerequisite mapping against WBCHSE, CBSE and ISC*

**Part III – The Finalised First-Year Structure and the Group Rotation** — *The two semester tables and how the science cycle rotates*

**Part IV – Detailed Syllabi: Theory Courses** — *Nine theory courses*

**Part V – Detailed Syllabi: Laboratory and Practical Courses** — *Eight laboratory and practical courses*

**Part VI – Detailed Syllabi: Sessional and Non-Credit Mandatory Courses** — *Four courses*

**Part VII – Sources and Benchmarking** — *Model curricula and institutional syllabi consulted*

## Foreword

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This document contains the detailed syllabus of the courses of the first year, prepared against the finalised NEP 2020 compliant first-year structure of the University (B.Tech. First Year, MAKAUT 2026), effective from the academic year 2026–27. Because Semesters I and II are common to every B.Tech. programme of the University, this syllabus is a University document rather than a departmental one.

**The group rotation.** The structure balances the load on the science and workshop faculty of the affiliated colleges by rotating the basic-science cycle across the two semesters. In the finalised structure, Physics is taken in the first semester by the Computer Science and its allied branches, and Chemistry in the first semester by other branches; the two exchange in the second semester. Engineering Graphics and Workshop Practice rotate in the same way, and the two mandatory non-credit courses — the Indian Constitution and the Indian Knowledge System — are likewise taken one in each semester and rotated by group. The content of each course is identical whichever semester it falls in, and this document therefore gives one syllabus for each course, marked “First or Second semester (by group)” where the rotation applies. Part V sets the rotation out in full.

The Programme Outcomes have been reduced from twelve to the eleven of the National Board of Accreditation’s GAPC v4.0 framework, aligned with the Washington Accord; every CO–PO articulation matrix in this document is accordingly an eleven-column matrix, and Part I sets out the eleven outcomes and the mapping from the former twelve. Second, Basic Electrical and Electronics Engineering is now a four-credit course delivered as 3-1-0 — three lectures and a tutorial — and its syllabus carries the strengthened safety, semiconductor and operational-amplifier content settled with the Board, its laboratory being modified to match. Third, Programming for Problem Solving now carries an introduction to Artificial Intelligence (a full module in the theory and two AI-tools exercises in the laboratory), and both are titled accordingly. Fourth, the contact hours of every course have been set against the University’s norm — twelve to fifteen hours for each credit of theory, and twenty-four to thirty hours for each credit of practical or sessional work — and the hours are shown in each course header. Fifth, Engineering Chemistry with its laboratory is a full part of the year through the rotation, the Mathematics courses are generic and branch-neutral, and Environmental Science carries the UGC draft Mission LiFE framework.

The first-semester courses are scoped a little lighter, because admission to the first year and the effective teaching window of the first semester is shorter than the calendar suggests. And the two Mathematics courses are generic: where a programme needs mathematics particular to it, that requirement is met in Mathematics-III in Semester III, which each Department frames, or in Module I of the relevant discipline course — so that the first-year mathematics, and the common examination in it, remain uniform across every branch.

## Part I — The Outcome-Based Framework

### I.1 The eleven Programme Outcomes (NBA GAPC v4.0)

The National Board of Accreditation, under its updated GAPC v4.0 framework aligned with the Washington Accord, defines eleven Programme Outcomes for undergraduate engineering, consolidating the former twelve by merging the societal and environmental attributes into a single outcome. These eleven are adopted as the Programme Outcomes of every B.Tech. programme of the University and are used in the articulation matrices of this document.

| PO   | Attribute                                  | What the graduate is expected to be able to do                                                                                                                        |
|------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PO1  | Engineering Knowledge                      | Apply mathematics, science, computing and engineering fundamentals to solve complex engineering problems.                                                             |
| PO2  | Problem Analysis                           | Identify, formulate and analyse complex engineering problems to reach substantiated conclusions.                                                                      |
| PO3  | Design / Development of Solutions          | Design systems, components or processes that meet specified needs with due regard for public safety and cultural, societal and environmental factors.                 |
| PO4  | Conduct Investigations of Complex Problems | Use research-based methods — design of experiment, analysis and interpretation of data — to reach valid conclusions.                                                  |
| PO5  | Engineering Tool Usage                     | Select and apply modern engineering and information-technology tools to complex activities, with an understanding of their limitations.                               |
| PO6  | The Engineer and the World                 | Assess the global, societal, health, safety, legal, cultural and environmental impacts of engineering practice (the former society and environment outcomes, merged). |
| PO7  | Ethics                                     | Apply professional ethics, human values, and the principles of diversity and inclusion.                                                                               |
| PO8  | Individual and Collaborative Team Work     | Function effectively as an individual and as a member or leader of diverse teams.                                                                                     |
| PO9  | Communication                              | Communicate clearly and inclusively with the engineering community and with society.                                                                                  |
| PO10 | Project Management and Finance             | Apply engineering-management and economic decision-making principles to projects.                                                                                     |
| PO11 | Life-Long Learning                         | Recognise the need for, and have the ability to engage in, independent and life-long learning.                                                                        |

*Programme Specific Outcomes are framed by each Department for its own programme; since these courses are common to all programmes, no PSO mapping is given here.*

### I.2 Mapping from the former twelve outcomes

For the reader familiar with the earlier twelve-outcome model, the correspondence is as follows, so that an existing mapping can be migrated without loss.

| Former (12 PO) model                                              | New (11 PO) model — GAPC v4.0                                    |
|-------------------------------------------------------------------|------------------------------------------------------------------|
| PO1 Engineering Knowledge                                         | PO1 Engineering Knowledge                                        |
| PO2 Problem Analysis                                              | PO2 Problem Analysis                                             |
| PO3 Design / Development of Solutions                             | PO3 Design / Development of Solutions                            |
| PO4 Conduct Investigations                                        | PO4 Conduct Investigations of Complex Problems                   |
| PO5 Modern Tool Usage                                             | PO5 Engineering Tool Usage                                       |
| PO6 The Engineer and Society + PO7 Environment and Sustainability | PO6 The Engineer and the World (the two merged)                  |
| PO8 Ethics                                                        | PO7 Ethics (now including human values, diversity and inclusion) |
| PO9 Individual and Team Work                                      | PO8 Individual and Collaborative Team Work                       |
| PO10 Communication                                                | PO9 Communication                                                |
| PO11 Project Management and Finance                               | PO10 Project Management and Finance                              |
| PO12 Life-Long Learning                                           | PO11 Life-Long Learning                                          |

### I.3 Levels of the revised Bloom taxonomy

| Level           | Verb family                  | What the student is asked to do in the first year                                                    |
|-----------------|------------------------------|------------------------------------------------------------------------------------------------------|
| L1 – Remember   | define, state, list, recall  | Reproduce a definition, a law or a standard value. Group A of the paper and the online test.         |
| L2 – Understand | explain, describe, classify  | Explain a principle or interpret a diagram or curve. Groups A and B.                                 |
| L3 – Apply      | compute, solve, use          | Work a problem or a procedure by an established method. Groups B and C, and the practical work.      |
| L4 – Analyse    | derive, compare, distinguish | Break a problem into parts, derive a result, or compare alternatives. Group C.                       |
| L5 – Evaluate   | assess, justify, select      | Choose between options against criteria and defend the choice. Group C and the case-based courses.   |
| L6 – Create     | design, formulate, construct | Reached in the first year chiefly in the design and modelling of Graphics and in later project work. |

### I.4 The articulation convention and the computation of attainment

The correlation between a Course Outcome and a Programme Outcome is recorded at three levels — 3 substantial, 2 moderate, 1 slight, and a dash for none — the mapping being at the level of the Course Outcome, so that attainment can be traced to an examination question or a laboratory task. A level of 3 is entered only where a question or task in this course could by itself furnish evidence of that outcome.

Attainment is computed for each Course Outcome from direct evidence (the marks on the questions and tasks mapped to it) and indirect evidence (an outcome-based student feedback instrument) in the ratio of eighty to twenty.

| Attainment level          | Criterion (direct)                                                                      | Action required                                                               |
|---------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Level 3 — attained        | Not less than 70 % of the class secures 60 % or more of the marks mapped to the outcome | Record and maintain.                                                          |
| Level 2 — partly attained | Between 55 % and 70 % of the class does so                                              | The teacher records a corrective action.                                      |
| Level 1 — not attained    | Fewer than 55 % of the class does so                                                    | The Department reviews delivery, prerequisite provision and question setting. |

The thresholds are indicative and are placed before the Board for adoption; whichever are adopted must be applied uniformly, since the online end-semester continuous assessment is common to all colleges.

## Part II — The Bridge from Class XII

The largest single cause of first-year failure is the discontinuity between what the school syllabus taught and what the first-year course assumes. This Part states the assumption course by course and provides for the gap.

| Course                                   | Assumed from the school syllabus                                                               | Board reference                          | Provision where absent                                                                                  |
|------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Mathematics I & II                       | Calculus, matrices, vectors, complex numbers, first-order ODEs                                 | CBSE XII Units II–IV, IX; WBCHSE; ISC    | Ten-hour non-credit bridge in the first three weeks, recorded in the WCA                                |
| Physics                                  | Oscillations and waves; optics; electricity and magnetism; modern physics                      | CBSE XI–XII; WBCHSE; ISC                 | Module I begins from the differential equation of SHM; earlier material revised in two hours            |
| Chemistry                                | Atomic structure and bonding; thermodynamics, equilibrium, electrochemistry; organic chemistry | CBSE XI–XII; WBCHSE; ISC                 | Three-hour bridge for students without Class XII Chemistry                                              |
| Biology for Engineers                    | General science of Class X; Higher Secondary Biology where taken                               | WBBSE / CBSE / ICSE X; CBSE XII Biology  | Module I is self-contained and supplies what a student without Class XII Biology requires               |
| Basic Electrical & Electronics           | Ohm's and Kirchhoff's laws; AC and the LCR circuit; diode, transistor and gates                | CBSE XII Units II–IV, IX–XI; WBCHSE; ISC | Mesh and nodal methods taught from first principles; safety and wiring assume nothing                   |
| Programming & Introduction to AI         | Computer Science / Informatics Practices where taken; otherwise nothing                        | CBSE XI–XII; WBCHSE Computer Application | No programming or AI assumed; Modules I–II compressed where the class has programmed                    |
| English / Language Lab                   | English of the Class XII course                                                                | WBCHSE / CBSE / ISC English              | A first-week diagnostic places students in graded groups                                                |
| Graphics & Workshop                      | Plane and solid geometry; 3-D coordinate geometry                                              | WBBSE / CBSE / ICSE; CBSE XII Unit IV    | No prior drawing or workshop experience assumed; each begins from the standard and the safety induction |
| Environmental Science; IKS; Constitution | General science, geography and civics of Class X                                               | WBBSE / CBSE / ICSE                      | Self-contained; no prior study, and no Sanskrit, assumed                                                |

**Three provisions apply across the first year.** The Mathematics bridge is conducted for the whole class; the English diagnostic is used for placement and not for marks; and the Department is to identify the students at risk within the first three weeks and place them in the remedial and mentoring arrangement required under NAAC Criterion 2.2.

## Part III — The Finalised First-Year Structure and the Group Rotation

### III.1 Semester I — Total 20 credits

| Category                   | UPID                | Code                         | Course                                                                            | L – T – P | Credit     | Notes       | Course Type |
|----------------------------|---------------------|------------------------------|-----------------------------------------------------------------------------------|-----------|------------|-------------|-------------|
| Basic Science course       | 1100001             | 26-BS-M101                   | Mathematics – I                                                                   | 3 – 0 – 0 | 3          | Both groups | Theory      |
| Basic Science course       | 1100002/<br>1100003 | 26-BS-PH101 /<br>26-BS-CH101 | Physics (CSE & allied) / Chemistry (other branches)                               | 3 – 0 – 0 | 3          | By group    | Theory      |
| Engineering Sc Course      | 1100004             | 26-ES-EE101                  | Basic Electrical and Electronics Engineering                                      | 3 – 1 – 0 | 4          | Both groups | Theory      |
| Ability Enhancement Course | 1100005             | 26-AEC-101                   | English for Communication Skills                                                  | 2 – 0 – 0 | 2          | Both groups | Theory      |
| Basic Science course       | 1100006/<br>1100007 | 26-BS-PH191 /<br>26-BS-CH191 | Physics Lab (CSE & allied) / Chemistry Lab (other branches)                       | 0 – 0 – 3 | 1.5        | By group    | Practical   |
| Engineering Science Course | 1100008/<br>1100009 | 26-ES-ME191 /<br>26-ES-ME192 | Engineering Graphics & Design (CSE & allied) / Workshop Practice (other branches) | 0 – 0 – 4 | 2          | By group    | Practical   |
| Engineering Science Course | 1100010             | 26-ES-EE191                  | Basic Electrical and Electronics Engineering Lab                                  | 0 – 0 – 3 | 1.5        | Both groups | Practical   |
| Value Added Course         | 1100011             | 26-VAC-181                   | NCC / NSS / Yoga and Sports                                                       | 0 – 0 – 4 | 2          | Both groups | Sessional   |
| Skill Enhancement Course   | 1100012             | 26-SEC-181                   | Skill Enhancement Course — MOOCs / Skills                                         | 0 – 0 – 2 | 1          | Both groups | Sessional   |
| Value Added Course         | 1100013/<br>1100014 | 26-MC-181 /<br>26-MC-182     | Indian Constitution (CSE & allied) / Indian Knowledge System (others)             | 2 – 0 – 0 | Non-credit | By group    | Sessional   |
|                            |                     |                              | Total of Semester I                                                               |           | 20         |             |             |

*Preceded by the three-week Induction Programme with Universal Human Values (mandatory, non-credit).*

### III.2 How the rotation is read

|             | Group A — CSE and allied branches                                                 | Group B — other branches                                                          |
|-------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Semester I  | Physics + Physics Lab; Engineering Graphics; Indian Constitution (non-credit)     | Chemistry + Chemistry Lab; Workshop Practice; Indian Knowledge System(non-credit) |
| Semester II | Chemistry + Chemistry Lab; Workshop Practice; Indian Knowledge System(non-credit) | Physics + Physics Lab; Engineering Graphics; Indian Constitution (non-credit)     |

Whichever semester a course falls in, its syllabus is identical; the detailed syllabi that follow are given once per course. The Department assigns its branches to the two groups so that the numbers, and hence the load on the science and workshop faculty, are balanced across the two semesters.

| Group A — CSE and allied branches                                                                                   | Group B — other branches                                    |
|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| B.Tech in Computer Sc. & Engineering                                                                                | B.Tech in Electronics & Communications Engg                 |
| B.Tech in Computer Science And Engineering (Artificial Intelligence and Machine Learning)                           | B.Tech in Electrical Engineering                            |
| B.Tech in Computer Science and Business System                                                                      | B.Tech in Mechanical Engineering                            |
| B.Tech in Computer Science and Engineering (Data Science)                                                           | B.Tech in Civil Engineering                                 |
| B.Tech in Computer Science and Engineering (Cyber Security)                                                         | B.Tech in Information Technology                            |
| B.Tech in Computer Science And Engineering (Internet of Things)                                                     | B.Tech in Electrical & Electronics Engineering              |
| B.Tech in Computer Science and Engineering (Artificial Intelligence)                                                | B.Tech in Textile Technology                                |
| B.Tech in Computer Science And Engineering (Internet of Things and Cyber Security including Block Chain Technology) | B.Tech in Applied Electronics & Instrumentation Engineering |
| B.Tech in Computer Science & Design (CSD)                                                                           | B.Tech in Food Technology                                   |
| B.Tech in Computer Science and Information Technology                                                               | B.Tech in Apparel & Production Management                   |
| B.Tech in Electronics and Computer Science                                                                          | B.Tech in Civil & Environmental Engineering                 |
| B.Tech in Electrical & Computer Engineering                                                                         | B.Tech in Biomedical Engineering                            |
| B.Tech in Artificial Intelligence and Machine Learning                                                              | B.Tech in Chemical Engineering                              |
| B.Tech in Artificial Intelligence And Data Science                                                                  | B.Tech in Leather Technology                                |
| B.Tech in Artificial Intelligence                                                                                   | B.Tech in Automobile Engineering                            |

## Part IV — Detailed Syllabi: Theory Courses

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The four theory courses follow, in the order in which they appear in the structure. Each carries a header block, prerequisites mapped to the school syllabus, objectives, Course Outcomes with Bloom levels, the eleven-column CO–PO matrix, the module-wise course plan with hours, cross-cutting notes, texts, suggested SWAYAM / NPTEL resources, the assessment map, and question-paper notes.

## 26-BS-M101 — Mathematics – I

|                     |                 |                      |                        |
|---------------------|-----------------|----------------------|------------------------|
| <b>Course Code</b>  | 26-BS-M101      | <b>Category</b>      | Basic Science Course   |
| <b>Course Title</b> | Mathematics – I | <b>L – T – P</b>     | 3 – 0 – 0              |
| <b>Credits</b>      | 3               | <b>Contact Hours</b> | 36                     |
| <b>Semester</b>     | First           | <b>Assessment</b>    | Theory: CA 30 + ESE 70 |

### Prerequisites

The course builds on the Higher Secondary Mathematics of WBCHSE, CBSE or ISC. Being a first-semester course it is scoped a little lighter, since admission to the first year is habitually late and the effective teaching window is shorter than the calendar suggests. The following are assumed:

- Differentiation and integration of elementary functions; the idea of a limit and of continuity (CBSE XII Units III–IV; WBCHSE Class XII; ISC).
- Matrices and determinants to the third order and the solution of a linear system (CBSE XII Unit III).
- Vectors in two and three dimensions (CBSE XII Unit IV) and complex numbers in algebraic and polar form (CBSE XI).
- A ten-hour bridge is conducted for the whole class in the first three weeks, re-establishing differentiation, integration and the standard notation; it is recorded in the whole-semester component and is not separately credited.

### Course Objectives

- To establish the differential and integral calculus of one and several variables as the working language of engineering analysis.
- To develop competence in the matrix and eigenvalue methods that underlie later courses across every branch.
- To introduce the convergence of sequences and series as the basis of approximation.
- To do so on a common, branch-neutral footing, leaving discipline-specific mathematics to Semester III.

### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                                                                  | Bloom        |
|-----|----------------------------------------------------------------------------------------------------------------------------|--------------|
| CO1 | Apply the differential calculus of one and several variables, including partial differentiation, to engineering functions. | L3 – Apply   |
| CO2 | Evaluate definite, improper and multiple integrals and use them to compute area, volume and average value.                 | L3 – Apply   |
| CO3 | Apply matrix methods — rank, the solution of linear systems, eigenvalues and eigenvectors — to engineering problems.       | L3 – Apply   |
| CO4 | Test an infinite series for convergence and construct elementary power-series approximations.                              | L4 – Analyse |

### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | 3   | 2   | 1   | 1   | 1   | –   | –   | –   | –   | –    | 1    |
| CO2     | 3   | 2   | 1   | 1   | 1   | –   | –   | –   | –   | –    | 1    |
| CO3     | 3   | 3   | 2   | 1   | 2   | –   | –   | –   | –   | –    | 1    |
| CO4     | 3   | 2   | 1   | 1   | 1   | –   | –   | –   | –   | –    | 2    |

## Course Plan — Module-wise Syllabus

| Module | Title                         | Contents                                                                                                                                                                                                                                                                                                                                 | Hours |
|--------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| I      | Calculus of a Single Variable | Review in brief of limits, continuity and differentiation; successive differentiation and Leibnitz's theorem; Rolle's and the mean-value theorems; Taylor's and Maclaurin's series with elementary applications; maxima and minima of a function of one variable; definite and improper integrals and the reduction formulae in outline. | 9     |
| II     | Calculus of Several Variables | Functions of two and more variables; partial derivatives; the total differential and its use in error estimation; the chain rule; Euler's theorem on homogeneous functions; maxima and minima of a function of two variables and the method of Lagrange multipliers in outline.                                                          | 9     |
| III    | Multiple Integrals            | Double integrals in Cartesian and polar coordinates; change of order of integration; computation of area and volume; the idea of the triple integral.                                                                                                                                                                                    | 8     |
| IV     | Matrices and Linear Algebra   | Rank of a matrix by elementary transformation; consistency and solution of a linear system; eigenvalues and eigenvectors; the Cayley–Hamilton theorem and its use; diagonalisation in outline. The linear-algebra content is generic; discipline-specific extensions are left to Mathematics-III.                                        | 10    |

**Cross-cutting notes.** This is a generic mathematics course, common to every branch of the University; it deliberately carries no discipline-specific material. Where a programme requires mathematics particular to it — discrete structures and graph theory for the computing branches, additional transforms and complex analysis for the electronics and electrical branches, additional statistics and numerical methods for the others — that requirement is to be met either in Mathematics-III in Semester III, which each Department frames for its own programme, or in Module I of the relevant discipline course where it is small enough to be carried there. The first-year mathematics is thus kept uniform, so that the common first-year examination and inter-branch mobility are preserved.

### Text and Reference Books

- Reena Garg, Mathematics - I, Calculus and Linear Algebra, Khanna Publishing House (AICTE Prescribed Textbook)
- Reena Garg, Advanced Engineering Mathematics, Khanna Publishing House
- Erwin Kreyszig, Advanced Engineering Mathematics, Wiley.
- G. B. Thomas and R. L. Finney, Calculus and Analytic Geometry, Pearson.
- B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers.
- N. P. Bali and M. Goyal, A Textbook of Engineering Mathematics, Laxmi Publications.

### Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Engineering Mathematics I (Prof. Jitendra Kumar, IIT Kharagpur), 12 weeks.
- NPTEL — Multivariable Calculus, 12 weeks.
- NPTEL — Matrix Analysis with Applications / Linear Algebra, 12 weeks.
- SWAYAM — Mathematics I (AICTE / SWAYAM), as offered.

**Question-paper note.** The paper is predominantly a problem paper: at least two-thirds of the marks of Groups B and C are for the working of problems, not for statement of theory. No question draws on discipline-specific material, since the class is common to all branches.

## 26-BS-PH101 — Physics

|                     |                            |                      |                        |
|---------------------|----------------------------|----------------------|------------------------|
| <b>Course Code</b>  | 26-BS-PH101                | <b>Category</b>      | Basic Science Course   |
| <b>Course Title</b> | Physics                    | <b>L – T – P</b>     | 3 – 0 – 0              |
| <b>Credits</b>      | 3                          | <b>Contact Hours</b> | 36                     |
| <b>Semester</b>     | First or Second (by group) | <b>Assessment</b>    | Theory: CA 30 + ESE 70 |

### Prerequisites

Builds on the Higher Secondary Physics of WBCHSE, CBSE or ISC. Under the group rotation it is taken in one semester by one group and in the other by the second; the syllabus is identical. Assumed:

- Oscillations, waves and the elements of optics (CBSE XI Units VII, X; WBCHSE; ISC).
- Electrostatics, current electricity and magnetism (CBSE XII Units I–IV).
- The elements of modern physics — the photoelectric effect and the Bohr atom (CBSE XII Units VIII–IX).
- Calculus to the level of Mathematics-I, taken concurrently.

### Course Objectives

- To establish the physics of oscillations and waves as the basis of much of engineering.
- To develop the electromagnetic theory and the wave picture of light used across the branches.
- To introduce quantum and solid-state physics as the foundation of modern devices.
- To connect each principle to an engineering application and to laboratory measurement.

### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                                                         | Bloom        |
|-----|-------------------------------------------------------------------------------------------------------------------|--------------|
| CO1 | Apply the theory of simple harmonic, damped and forced oscillations and of waves to physical systems.             | L3 – Apply   |
| CO2 | Apply the principles of interference, diffraction and polarisation to optical phenomena and instruments.          | L3 – Apply   |
| CO3 | Apply the laws of electromagnetism, including Maxwell's equations in outline, to fields and waves.                | L3 – Apply   |
| CO4 | Apply the elements of quantum and solid-state physics to the behaviour of materials and devices.                  | L4 – Analyse |
| CO5 | Relate the theory to measurement, estimating the uncertainty of a physical quantity determined in the laboratory. | L4 – Analyse |

### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | 3   | 2   | 1   | 1   | 1   | –   | –   | –   | –   | –    | 1    |
| CO2     | 3   | 2   | 1   | 2   | 1   | –   | –   | –   | –   | –    | 1    |
| CO3     | 3   | 3   | 2   | 2   | 2   | 1   | 1   | –   | –   | –    | 1    |
| CO4     | 3   | 2   | 2   | 2   | 2   | 1   | 1   | –   | –   | –    | 2    |
| CO5     | 3   | 2   | 1   | 3   | 2   | –   | –   | 1   | 1   | –    | 2    |

## Course Plan — Module-wise Syllabus

| Module | Title                          | Contents                                                                                                                                                                                                                                                            | Hours |
|--------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| I      | Oscillations and Waves         | Simple harmonic motion from the differential equation; damped and forced oscillations, resonance and the quality factor; coupled oscillators in outline; the wave equation, travelling and standing waves; superposition, beats and the Doppler effect.             | 8     |
| II     | Wave Optics                    | Interference — Young's experiment, thin films, Newton's rings; diffraction — single slit, the grating and its resolving power; polarisation — Malus' law, double refraction, the quarter- and half-wave plate; the elements of the laser and of optical fibre.      | 8     |
| III    | Electromagnetism               | Electrostatics and magnetostatics in review; the laws of Gauss, Faraday and Ampère; the displacement current; Maxwell's equations in integral and differential form (statement and meaning); the electromagnetic wave in vacuum in outline.                         | 8     |
| IV     | Quantum Physics                | The failure of classical physics; the photoelectric and Compton effects; the de Broglie hypothesis and wave–particle duality; the uncertainty principle; the Schrödinger equation and the particle in a box; the quantum numbers in outline.                        | 7     |
| V      | Solid-State and Modern Physics | Free-electron theory of metals in outline; the band theory of solids and the distinction of conductor, semiconductor and insulator; intrinsic and extrinsic semiconductors; superconductivity in brief; an introduction to nanomaterials and their engineering use. | 5     |

**Cross-cutting notes.** The physics laboratory (BS-PH191/291) is taken concurrently and is assessed separately; the theory is to be sequenced so that the relevant principle precedes the experiment. Module V connects forward to the Basic Electrical and Electronics course and to the materials content of the disciplines.

### Text and Reference Books

- A. B. Bhattacharya, Introduction to Electromagnetic Theory (with Lab Manual), Khanna Publishing House (AICTE Prescribed Textbook)
- A. Beiser, Concepts of Modern Physics, McGraw-Hill.
- D. J. Griffiths, Introduction to Electrodynamics, Pearson.
- Halliday, Resnick and Walker, Fundamentals of Physics, Wiley.
- A. B. Bhattacharya, Engineering Physics, Khanna Publishing House
- B. B. Laud, Electromagnetics, New Age.
- A. Ghatak, Optics, McGraw-Hill.

### Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Engineering Physics I / II (IITs), 12 weeks.
- NPTEL — Waves and Oscillations, 8 / 12 weeks.
- NPTEL — Semiconductor Physics and Devices, 12 weeks.
- Virtual Labs (Ministry of Education) — Optics, Modern Physics and Electricity & Magnetism laboratories.

**Question-paper note.** At least one question in Group C is to be a numerical problem — a resonance, interference-fringe, field or quantum calculation — with units and significant figures required. Standard constants are printed with the paper.

## 26-BS-CH101 — Chemistry

|                     |                            |                      |                        |
|---------------------|----------------------------|----------------------|------------------------|
| <b>Course Code</b>  | 26-BS-CH101                | <b>Category</b>      | Basic Science Course   |
| <b>Course Title</b> | Chemistry                  | <b>L – T – P</b>     | 3 – 0 – 0              |
| <b>Credits</b>      | 3                          | <b>Contact Hours</b> | 36                     |
| <b>Semester</b>     | First or Second (by group) | <b>Assessment</b>    | Theory: CA 30 + ESE 70 |

### Prerequisites

Builds on the Higher Secondary Chemistry of WBCHSE, CBSE or ISC. Under the group rotation it is taken in one semester by one group and in the other by the second; the syllabus is identical. Assumed:

- Atomic structure, the quantum numbers and electronic configuration; the periodic table (CBSE XI Units II–III; WBCHSE).
- Chemical bonding — ionic, covalent and coordinate bonds, VSEPR and hybridisation (CBSE XI Unit IV).
- Thermodynamics, equilibrium, electrochemistry and kinetics (CBSE XI Unit VI; XII Units III–IV).
- A three-hour bridge on atomic structure, bonding and the mole concept is provided for students without Class XII Chemistry.

### Course Objectives

- To establish the quantum-mechanical picture of the atom and the molecule as the basis of the properties of materials.
- To develop the thermodynamic, kinetic and electrochemical reasoning used to predict whether and how fast a process occurs.
- To connect the structure of materials and of water to their engineering behaviour and treatment.
- To introduce the instrumental methods by which composition and structure are determined.

### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                                                                       | Bloom        |
|-----|---------------------------------------------------------------------------------------------------------------------------------|--------------|
| CO1 | Apply the quantum-mechanical and molecular-orbital picture to the structure and properties of molecules and materials.          | L3 – Apply   |
| CO2 | Apply thermodynamics, chemical and phase equilibrium and reaction kinetics to engineering processes and compute the quantities. | L3 – Apply   |
| CO3 | Apply the principles of electrochemistry to cells, batteries and fuel cells, and explain and control corrosion.                 | L3 – Apply   |
| CO4 | Relate the structure of engineering materials and the chemistry of water to their engineering use and treatment.                | L4 – Analyse |
| CO5 | Select an appropriate spectroscopic or instrumental method for an analytical problem and interpret its output.                  | L3 – Apply   |

### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | 3   | 3   | 1   | 1   | 1   | –   | –   | –   | –   | –    | 1    |
| CO2     | 3   | 3   | 2   | 2   | 1   | –   | –   | –   | –   | –    | 1    |
| CO3     | 3   | 2   | 2   | 2   | 2   | 1   | 2   | –   | –   | –    | 2    |
| CO4     | 3   | 2   | 2   | 2   | 2   | 2   | 2   | –   | –   | –    | 2    |
| CO5     | 3   | 2   | 1   | 3   | 2   | –   | –   | –   | –   | –    | 2    |

## Course Plan — Module-wise Syllabus

| Module | Title                                    | Contents                                                                                                                                                                                                                                                                                                                                  | Hours |
|--------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| I      | Atomic and Molecular Structure           | The Schrödinger equation and the particle in a box as a model; atomic and molecular orbitals; molecular-orbital theory of diatomic molecules, bond order and magnetic behaviour; $\pi$ -molecular orbitals of conjugated systems; band theory of solids in outline; crystal-field splitting and the colour of transition-metal complexes. | 8     |
| II     | Thermodynamics, Equilibrium and Kinetics | The laws of thermodynamics in review; entropy, free energy and the criteria of spontaneity; chemical potential; chemical and phase equilibrium, the phase rule and simple phase diagrams; chemical kinetics — order, molecularity, the rate law, the Arrhenius equation and catalysis.                                                    | 8     |
| III    | Electrochemistry and Corrosion           | Electrode potential and the electrochemical series; the Nernst equation; galvanic and concentration cells; conductance; batteries — primary, secondary and the lithium-ion cell — and fuel cells; corrosion — mechanism, factors, and protection by coatings, cathodic protection and inhibitors.                                         | 7     |
| IV     | Engineering Materials and Water          | Polymers — addition and condensation polymerisation, thermoplastics and thermosets, elastomers, conducting and biodegradable polymers; composites and nanomaterials in outline. Water — hardness and its determination, boiler troubles, softening by lime-soda, zeolite and ion-exchange methods, and municipal treatment.               | 8     |
| V      | Instrumental Methods of Analysis         | The basis of spectroscopy and the Beer–Lambert law; ultraviolet–visible, infrared and, in outline, nuclear magnetic resonance spectroscopy; elementary chromatography; the accuracy and precision of a measurement and the reporting of a result with its uncertainty.                                                                    | 5     |

**Cross-cutting notes.** Module III (corrosion) and Module IV (materials and water) are the points at which the course serves the materials, durability and water concerns of the Civil, Mechanical, Electrical and Leather disciplines and connects to Environmental Science. The concurrent Chemistry Laboratory (BS-CH191/291) reduces at least one result with its uncertainty.

### Text and Reference Books

- Manisha Aggrawal, Chemistry - I, Khanna Publishing House (AICTE Prescribed Textbook)
- P. C. Jain and M. Jain, Engineering Chemistry, Dhanpat Rai.
- P. W. Atkins, Physical Chemistry, Oxford University Press.
- Manisha Aggrawal, Engineering Chemistry, Khanna Publishing House
- B. R. Puri, L. R. Sharma and M. S. Pathania, Principles of Physical Chemistry, Vishal.
- S. S. Dara and S. S. Umare, A Textbook of Engineering Chemistry, S. Chand.
- Shashi Chawla, A Textbook of Engineering Chemistry, Dhanpat Rai.

### Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Chemistry: Principles and Applications / Engineering Chemistry, 12 weeks.
- NPTEL — Electrochemical Energy Systems / Corrosion Engineering, 12 weeks.
- Virtual Labs (Ministry of Education) — Physical, Inorganic and Analytical Chemistry laboratories.

**Question-paper note.** At least one Group C question is a numerical problem — a free-energy or equilibrium computation, a Nernst potential, a rate or activation-energy calculation, or a hardness determination. Structure-and-reasoning questions require explanation from the underlying bonding or thermodynamics.

## 26-ES-EE101 — Basic Electrical and Electronics Engineering

|                     |                                              |                      |                            |
|---------------------|----------------------------------------------|----------------------|----------------------------|
| <b>Course Code</b>  | 26-ES-EE101                                  | <b>Category</b>      | Engineering Science Course |
| <b>Course Title</b> | Basic Electrical and Electronics Engineering | <b>L – T – P</b>     | 3 – 1 – 0                  |
| <b>Credits</b>      | 4                                            | <b>Contact Hours</b> | 44                         |
| <b>Semester</b>     | First                                        | <b>Assessment</b>    | Theory: CA 30 + ESE 70     |

### Prerequisites

A four-credit course (3-1-0), the tutorial hour serving the problem-working that a combined electrical-and-electronics paper needs. Scoped at about forty-four hours. Assumed:

- Ohm's and Kirchhoff's laws and simple resistive circuits (CBSE XII Unit II; WBCHSE).
- Alternating current and the LCR circuit in outline (CBSE XII Unit IV).
- The p–n junction diode and the elements of the transistor and logic gates (CBSE XII Units IX–XI).
- The calculus of Mathematics-I, taken concurrently.

### Course Objectives

- To establish the analysis of direct- and alternating-current circuits by the network theorems.
- To introduce electrical machines and their engineering use.
- To establish the physics and characteristics of the semiconductor devices — diode, transistor, field-effect transistor and operational amplifier.
- To ground the student in electrical safety and the Indian wiring practice.

### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                                                                            | Bloom           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| CO1 | Apply electrical safety and wiring practice and analyse direct-current circuits using the network theorems.                          | L3 – Apply      |
| CO2 | Analyse single-phase and balanced three-phase alternating-current circuits and compute power and power factor.                       | L3 – Apply      |
| CO3 | Explain the construction and characteristics of transformers and rotating machines and their engineering use.                        | L2 – Understand |
| CO4 | Explain the physics and characteristics of the semiconductor diode and its applications in rectification and regulation.             | L3 – Apply      |
| CO5 | Calculate the voltage gain of the operational amplifier in the inverting and non-inverting configuration and study its applications. | L3 – Apply      |

### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | 3   | 3   | 2   | 2   | 2   | 1   | 1   | 1   | –   | –    | 1    |
| CO2     | 3   | 3   | 2   | 2   | 2   | –   | –   | –   | –   | –    | 1    |
| CO3     | 3   | 2   | 2   | 1   | 2   | 1   | 1   | –   | –   | –    | 1    |
| CO4     | 3   | 3   | 2   | 2   | 2   | –   | –   | –   | –   | –    | 2    |
| CO5     | 3   | 3   | 3   | 2   | 3   | –   | –   | 1   | –   | –    | 2    |

## Course Plan — Module-wise Syllabus

| Module | Title                                             | Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Hours |
|--------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| I      | Basics of Electrical Wiring and Safety            | Electrical safety — the effect of current on the body; earthing; the residual-current device; the Switch-Fuse Unit (SFU), MCB, ELCB and MCCB; types of wires and cables; the elements of the Indian wiring practice; types of batteries and their important characteristics.                                                                                                                                                                                                                                                                                      | 3     |
| II     | Direct-Current Circuits                           | Circuit elements and their voltage–current relations; independent and dependent sources; Kirchhoff's voltage and current laws; mesh and nodal analysis; star–delta transformation; superposition, Thevenin, Norton and maximum-power-transfer theorems; transient response of RL and RC circuits in outline.                                                                                                                                                                                                                                                      | 9     |
| III    | Alternating-Current Circuits                      | Generation of alternating voltage; average and root-mean-square values, form and peak factors; phasor representation; analysis of R, L, C and series and parallel RLC circuits; impedance and admittance; real, reactive and apparent power, power factor and its improvement; series and parallel resonance; balanced three-phase systems, star and delta connection, and three-phase power measurement in outline.                                                                                                                                              | 8     |
| IV     | Transformers and Rotating Machines                | Single-phase transformer — construction, principle, equivalent circuit, losses, efficiency and regulation; the auto-transformer in outline. Direct-current machines — construction, e.m.f. and torque equations, types of excitation, characteristics and starting. Introduction to alternating-current motors (three-phase and single-phase), with the stepper motor and the brushless direct-current motor and a note on their use in automation.                                                                                                               | 6     |
| V      | Semiconductor Fundamentals and Diode Applications | Energy bands in solids; valence and conduction bands; insulators, semiconductors and conductors; intrinsic and extrinsic semiconductors; current in semiconductors by diffusion and drift; the p–n junction diode under forward and reverse bias and its current–voltage characteristic; breakdown — Zener and avalanche; rectifiers — half-wave and full-wave with filters; the Zener diode as a voltage regulator; the light-emitting diode and the photodiode.                                                                                                 | 7     |
| VI     | Transistors and Operational Amplifiers            | Bipolar junction transistor — structure and symbol, operation, modes, the CB, CE and CC characteristics and current-gain expressions, the Early effect, and the hybrid (h-parameter) model in outline. Field-effect transistor — the junction FET: circuit, symbol, drain and transfer characteristics, and the saturation drain current; the MOSFET — construction, operation and drain characteristics. The operational amplifier — the ideal op-amp, open-loop characteristics and parameters; the inverting, non-inverting, summing and difference amplifier. | 11    |

**Note on scope and delivery.** The course carries four credits as 3-1-0 — three lectures and a tutorial — in recognition that it combines two subjects within one paper. The tutorial hour is for the working of circuit and device problems and is not a fourth lecture. The safety-and-wiring Module I and the semiconductor and operational-amplifier content of Modules V and VI are the parts strengthened at the Board's instance, so that the electronics half is given its due.

### Text and Reference Books

- S.K. Sahdev, Basic Electrical Engineering (with Lab Manual), Khanna Publishing House (AICTE Prescribed Textbook)
- Ritu Sahdev, Basic Electrical Engineering, Khanna Publishing House.
- D. P. Kothari and I. J. Nagrath, Basic Electrical and Electronics Engineering, McGraw-Hill.
- D. C. Kulshreshtha, Basic Electrical Engineering, McGraw-Hill.
- Boylestad and Nashelsky, Electronic Devices and Circuit Theory, Pearson.
- V. K. Mehta and Rohit Mehta, Principles of Electronics, S. Chand.
- Millman, Halkias and Jit, Electronic Devices and Circuits, McGraw-Hill.

### Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Basic Electrical Circuits / Basic Electrical Engineering (IITs), 12 weeks.
- NPTEL — Electronic Devices and Circuits, 12 weeks.
- NPTEL — Analog Electronic Circuits / Operational Amplifiers, 12 weeks.
- Virtual Labs (Ministry of Education) — Basic Electrical and Analog Electronics laboratories.

**Question-paper note.** The paper is a problem-and-reasoning paper. At least one Group C question requires network analysis by a stated theorem, one requires an op-amp gain calculation (CO5), and one requires interpretation of a device characteristic. Circuit diagrams are to be drawn to standard symbols.

## 26-AEC-101 — English for Communication Skills

|                     |                                  |                      |                            |
|---------------------|----------------------------------|----------------------|----------------------------|
| <b>Course Code</b>  | 26-AEC-101                       | <b>Category</b>      | Ability Enhancement Course |
| <b>Course Title</b> | English for Communication Skills | <b>L – T – P</b>     | 2 – 0 – 0                  |
| <b>Credits</b>      | 2                                | <b>Contact Hours</b> | 26                         |
| <b>Semester</b>     | First                            | <b>Assessment</b>    | Theory: CA 30 + ESE 70     |

### Prerequisites

A two-credit first-semester course scoped at about twenty-six hours, and lightened for the shorter first-semester window. A first-week diagnostic places students in graded groups. Assumed:

- The English of the Higher Secondary course (WBCHSE / CBSE / ISC).
- No uniform standard is assumed; the diagnostic test is used for placement and not for marks.
- The Language Laboratory (AEC-291, Semester II) carries the intensive listening-and-speaking practice.

### Course Objectives

- To strengthen the grammar and vocabulary of technical English.
- To develop reading, comprehension and the conventions of technical writing.
- To build the listening and speaking competence the workplace requires.
- To prepare the student for the communication demands of employability.

### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                               | Bloom      |
|-----|-----------------------------------------------------------------------------------------|------------|
| CO1 | Apply the grammar and vocabulary of technical English accurately in writing and speech. | L3 – Apply |
| CO2 | Read and comprehend a technical text and summarise it accurately.                       | L3 – Apply |
| CO3 | Write clear technical prose — the paragraph, the letter, the report and the e-mail.     | L3 – Apply |
| CO4 | Listen and speak effectively in professional and group settings.                        | L3 – Apply |

### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | 1   | 1   | –   | –   | 1   | 1   | 1   | 2   | 3   | –    | 2    |
| CO2     | 1   | 1   | –   | 1   | 1   | 1   | –   | 1   | 3   | –    | 2    |
| CO3     | 1   | 1   | 1   | 1   | 2   | 1   | 1   | 2   | 3   | 1    | 2    |
| CO4     | 1   | –   | –   | –   | 1   | 2   | 1   | 3   | 3   | –    | 2    |

## Course Plan — Module-wise Syllabus

| Module | Title                                           | Contents                                                                                                                                                                                                                                             | Hours |
|--------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| I      | The Grammar and Vocabulary of Technical English | Parts of speech and sentence structure in review; common errors and their correction; tense, voice and concord; the vocabulary of technical English — word-formation, technical and sub-technical vocabulary, and precision of usage.                | 6     |
| II     | Reading and Comprehension                       | Reading strategies — skimming, scanning and intensive reading; comprehension of a technical passage; note-making and summarising; interpreting data presented in a table or figure. (May be carried partly by guided assignment in a short session.) | 6     |
| III    | Technical Writing                               | The technical paragraph and its coherence; the formal letter and the e-mail; the memorandum; the structure of the technical report; the description of a process and of an object; the abstract.                                                     | 8     |
| IV     | Listening, Speaking and Employability           | Listening comprehension; pronunciation and stress in outline; the group discussion and the interview; the presentation; the résumé and the covering letter; communication for the workplace.                                                         | 6     |

**Cross-cutting notes.** The course is paired with the Language Laboratory (AEC-291) and with Essentials for Professional Skills (AEC-281), which carry the practice; this course supplies the frame. It serves PO9 (Communication) directly and is the principal first-year vehicle for it.

### Text and Reference Books

- Kulbhushan Kumar, English (with Lab Manual), Khanna Publishing House (AICTE Prescribed Textbook).
- Anjana Tiwari, Communication Skills in English, Khanna Publishing House (AICTE Prescribed Textbook).
- Meenakshi Raman and Sangeeta Sharma, Technical Communication, Oxford University Press.
- Board of Editors, English for Engineers and Technologists, Orient BlackSwan.
- F. T. Wood, Remedial English Grammar for Foreign Students, Macmillan.
- Kulbhushan Kumar, Effective Communication Skills, Khanna Publishing House.
- Ashraf Rizvi, Effective Technical Communication, McGraw-Hill.

### Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Technical English for Engineers / English Language for Competitive Examinations, 8 / 12 weeks.
- NPTEL — Effective Writing / Speaking Effectively, as offered.
- SWAYAM — Soft Skills and Communication, as offered.

**Question-paper note.** The paper tests use rather than the naming of rules: grammar is examined through correction and transformation, writing through a short composition, and reading through comprehension of an unseen technical passage.

## Part V — Detailed Syllabi: Laboratory and Practical Courses

The three laboratory and practical courses follow. Each carries its list of experiments or exercises mapped to the Course Outcomes, and is assessed under the practical scheme of Part III.

### 26-BS-PH191 — Physics Laboratory

|                     |                            |                      |                            |
|---------------------|----------------------------|----------------------|----------------------------|
| <b>Course Code</b>  | 26-BS-PH191                | <b>Category</b>      | Basic Science Course       |
| <b>Course Title</b> | Physics Laboratory         | <b>L – T – P</b>     | 0 – 0 – 3                  |
| <b>Credits</b>      | 1.5                        | <b>Contact Hours</b> | 36                         |
| <b>Semester</b>     | First or Second (by group) | <b>Assessment</b>    | Practical: PCA 40 + ESE 60 |

#### Prerequisites

Concurrent with Physics and sequenced so the relevant theory precedes the experiment. Assumed: the laboratory safety induction (first session); elementary measurement, significant figures and the reporting of a result with its uncertainty.

- The laboratory safety and measurement induction, conducted in the first session.
- Elementary use of the vernier, screw gauge and travelling microscope (Class XI–XII practical).
- The idea of least count, error and the reporting of a result with its uncertainty.

#### Course Objectives

- To connect the principles of the Physics course to measurement.
- To develop competence and safe practice in physical measurement.
- To establish the estimation of uncertainty as routine.
- To establish the discipline of the laboratory record.

#### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                                                       | Bloom        |
|-----|-----------------------------------------------------------------------------------------------------------------|--------------|
| CO1 | Work safely and take physical measurements to their rated precision using standard instruments.                 | L3 – Apply   |
| CO2 | Determine physical quantities — g, wavelength, refractive index, resistance, and the like — from measured data. | L3 – Apply   |
| CO3 | Plot and interpret an experimental graph and extract a physical quantity from its slope or intercept.           | L4 – Analyse |
| CO4 | Estimate the uncertainty in a measured result and maintain a reproducible laboratory record.                    | L4 – Analyse |

#### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | 3   | 2   | 1   | 3   | 2   | –   | –   | 2   | 1   | –    | 2    |
| CO2     | 3   | 3   | 2   | 3   | 2   | –   | –   | 2   | 1   | –    | 2    |
| CO3     | 3   | 3   | 1   | 3   | 3   | –   | –   | 2   | 2   | –    | 2    |
| CO4     | 2   | 2   | 1   | 3   | 2   | –   | –   | 3   | 2   | –    | 2    |

## List of Experiments / Exercises

| Sl. | Experiment / Exercise                                                                                             | CO            |
|-----|-------------------------------------------------------------------------------------------------------------------|---------------|
| 1   | Determination of the acceleration due to gravity by the bar or Kater's pendulum.                                  | CO1, CO2      |
| 2   | Determination of the spring constant and the acceleration due to gravity by a helical spring.                     | CO2, CO3      |
| 3   | Determination of the wavelength of light by Newton's rings.                                                       | CO2, CO3      |
| 4   | Determination of the wavelength of the spectral lines by a diffraction grating and the spectrometer.              | CO2, CO3      |
| 5   | Determination of the refractive index of the material of a prism by the spectrometer.                             | CO1, CO2      |
| 6   | Determination of an unknown resistance by the Carey Foster bridge.                                                | CO2, CO3      |
| 7   | Study of the characteristics of a p–n junction diode / the variation of a thermistor resistance with temperature. | CO2, CO3      |
| 8   | Determination of the specific resistance of a wire by the metre bridge.                                           | CO2, CO4      |
| 9   | Determination of the numerical aperture of an optical fibre / the wavelength of a laser by a grating.             | CO2, CO3      |
| 10  | Determination of the Planck constant using light-emitting diodes.                                                 | CO2, CO3, CO4 |

*A student is to perform not fewer than eight experiments. Every result is to be reported with its estimated uncertainty; a result reported without one is incomplete. At least one experiment is to be recorded and reduced using a spreadsheet or a short script.*

## Text and Reference Books

- A. B. Bhattacharya, Physics (Introduction to Electromagnetic Theory) (with Lab Manual), Khanna Publishing House.
- A. B. Bhattacharya, Physics (Oscillations, Waves and Optics) (with Lab Manual), Khanna Publishing House.
- C. L. Arora, Practical Physics, S. Chand.
- B. L. Worsnop and H. T. Flint, Advanced Practical Physics for Students, Methuen.
- J. R. Taylor, An Introduction to Error Analysis, University Science Books — for the uncertainty component.

## Suggested SWAYAM / NPTEL and Online Resources

- Virtual Labs (Ministry of Education) — Optics, Mechanics and Electricity & Magnetism laboratories, for pre-laboratory preparation.

## Indicative Assessment Map

| Component                               | Coverage                                                                                                                | Outcomes |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------|
| Practical CA – I (PCA1, 40)             | First cycle — conduct and safe working 10, observation and data 10, computation and result with uncertainty 10, viva 10 | CO1, CO2 |
| Practical CA – II (PCA2, 40)            | Second cycle, assessed on the same rubric                                                                               | CO3, CO4 |
| End-Semester Practical Examination (60) | Record, performance and viva-voce before an external examiner                                                           | All COs  |

## 26-BS-CH191 — Chemistry Laboratory

|                     |                            |                      |                            |
|---------------------|----------------------------|----------------------|----------------------------|
| <b>Course Code</b>  | 26-BS-CH191                | <b>Category</b>      | Basic Science Course       |
| <b>Course Title</b> | Chemistry Laboratory       | <b>L – T – P</b>     | 0 – 0 – 3                  |
| <b>Credits</b>      | 1.5                        | <b>Contact Hours</b> | 36                         |
| <b>Semester</b>     | First or Second (by group) | <b>Assessment</b>    | Practical: PCA 40 + ESE 60 |

### Prerequisites

Concurrent with Chemistry, sequenced so the relevant module precedes the experiment. Assumed: the laboratory safety induction (first session); elementary volumetric practice; significant figures and uncertainty.

- The laboratory safety induction, in the first session; no reagent is handled without it.
- Elementary use of the burette, pipette and volumetric flask (Class XI–XII practical).
- The reporting of a result with its unit and uncertainty.

### Course Objectives

- To develop competence and safe practice in quantitative chemical measurement.
- To connect the thermodynamic, kinetic, electrochemical and analytical principles of the theory to observation.
- To introduce instrumental measurement and the estimation of uncertainty as routine.
- To establish the discipline of the laboratory record.

### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                                                       | Bloom        |
|-----|-----------------------------------------------------------------------------------------------------------------|--------------|
| CO1 | Work safely in a chemical laboratory and carry out volumetric determinations to their rated precision.          | L3 – Apply   |
| CO2 | Determine physicochemical quantities — hardness, rate constant, cell potential, viscosity — from measured data. | L3 – Apply   |
| CO3 | Operate elementary instruments — pH meter, conductivity meter, colorimeter — and interpret their output.        | L3 – Apply   |
| CO4 | Estimate the uncertainty in a measured result and maintain a reproducible laboratory record.                    | L4 – Analyse |

### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | 3   | 2   | 1   | 3   | 2   | 1   | 2   | 2   | 1   | –    | 2    |
| CO2     | 3   | 3   | 2   | 3   | 2   | –   | 1   | 2   | 1   | –    | 2    |
| CO3     | 3   | 2   | 1   | 3   | 3   | –   | –   | 2   | 1   | –    | 2    |
| CO4     | 2   | 2   | 1   | 3   | 2   | –   | –   | 3   | 2   | –    | 2    |

## List of Experiments / Exercises

| Sl. | Experiment / Exercise                                                                                                                            | CO       |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1   | Laboratory safety, the handling of reagents and glassware, measurement, significant figures and uncertainty; preparation of a standard solution. | CO1, CO4 |
| 2   | Determination of the total, permanent and temporary hardness of water by the EDTA method.                                                        | CO1, CO2 |
| 3   | Estimation of a carbonate–bicarbonate mixture by double-indicator titration.                                                                     | CO1, CO2 |
| 4   | Determination of iron in a sample by a redox titration (permanganometry or dichromatometry).                                                     | CO1, CO2 |
| 5   | Determination of the rate constant of the acid-catalysed hydrolysis of an ester.                                                                 | CO2, CO4 |
| 6   | Determination of the strength of an acid by conductometric titration.                                                                            | CO2, CO3 |
| 7   | Determination of the strength or dissociation constant of a weak acid by pH-metric titration.                                                    | CO2, CO3 |
| 8   | Verification of the Beer–Lambert law and colorimetric determination of a coloured species.                                                       | CO2, CO3 |
| 9   | Determination of the viscosity of a liquid and its variation with concentration.                                                                 | CO2      |
| 10  | Determination of the cell constant and equivalent conductance of a strong electrolyte.                                                           | CO2, CO3 |
| 11  | Preparation of a simple polymer (phenol–formaldehyde or urea–formaldehyde) or of a metal complex.                                                | CO1      |

*A student is to perform not fewer than eight experiments. At least one experiment per cycle is recorded and reduced using a spreadsheet or short script, and every result is reported with its estimated uncertainty.*

## Text and Reference Books

- Manisha Aggrawal, Chemistry (with Lab Manual), Khanna Publishing House.
- B. Viswanathan and P. S. Raghavan, Practical Physical Chemistry, Viva Books.
- A. I. Vogel, A Textbook of Quantitative Chemical Analysis, Pearson.
- Sunita Rattan, Experiments in Applied / Engineering Chemistry, S. K. Kataria.

## Suggested SWAYAM / NPTEL and Online Resources

- Virtual Labs (Ministry of Education) — Physical, Analytical and Inorganic Chemistry laboratories.

## Indicative Assessment Map

| Component                               | Coverage                                                                                                                | Outcomes      |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------|
| Practical CA – I (PCA1, 40)             | First cycle — conduct and safe working 10, observation and data 10, computation and result with uncertainty 10, viva 10 | CO1, CO2      |
| Practical CA – II (PCA2, 40)            | Second cycle, assessed on the same rubric                                                                               | CO2, CO3, CO4 |
| End-Semester Practical Examination (60) | Record, performance and viva-voce before an external examiner                                                           | All COs       |

## 26-ES-EE191 — Basic Electrical and Electronics Engineering Laboratory

|                     |                                                         |                      |                            |
|---------------------|---------------------------------------------------------|----------------------|----------------------------|
| <b>Course Code</b>  | 26-ES-EE191                                             | <b>Category</b>      | Engineering Science Course |
| <b>Course Title</b> | Basic Electrical and Electronics Engineering Laboratory | <b>L – T – P</b>     | 0 – 0 – 3                  |
| <b>Credits</b>      | 1.5                                                     | <b>Contact Hours</b> | 36                         |
| <b>Semester</b>     | First                                                   | <b>Assessment</b>    | Practical: PCA 40 + ESE 60 |

### Prerequisites

Concurrent with Basic Electrical and Electronics Engineering (ES-EE101). Assumed: the electrical-safety induction (first session); the use of the multimeter, the cathode-ray oscilloscope and the function generator, taught in the first two sessions.

- The electrical-safety induction, in the first session.
- The use of the multimeter, oscilloscope, function generator and regulated power supply, taught at the outset.
- The circuit and device theory of ES-EE101, taken concurrently.

### Course Objectives

- To verify the network theorems and the behaviour of alternating-current circuits by measurement.
- To study the characteristics of electrical machines.
- To obtain and interpret the characteristics of the semiconductor devices and to test rectifier and regulator circuits.
- To study the operational amplifier in its basic configurations.

### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                                                                                                        | Bloom        |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| CO1 | Use electrical measuring instruments safely and verify the network theorems by measurement.                                                                      | L3 – Apply   |
| CO2 | Measure the parameters of an alternating-current circuit and study resonance and power-factor improvement.                                                       | L3 – Apply   |
| CO3 | Obtain and interpret the characteristics of the diode, the Zener diode, the bipolar junction transistor and the JFET, and test rectifier and regulator circuits. | L4 – Analyse |
| CO4 | Study the characteristics of an electrical machine or transformer.                                                                                               | L3 – Apply   |
| CO5 | Calculate the voltage gain of the operational amplifier in the inverting and non-inverting configuration.                                                        | L3 – Apply   |

### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | 3   | 3   | 2   | 3   | 3   | 1   | 1   | 2   | 1   | –    | 2    |
| CO2     | 3   | 3   | 2   | 3   | 3   | –   | –   | 2   | 1   | –    | 2    |
| CO3     | 3   | 3   | 2   | 3   | 3   | –   | –   | 2   | 1   | –    | 2    |
| CO4     | 3   | 2   | 2   | 3   | 3   | 1   | 1   | 2   | 1   | –    | 2    |
| CO5     | 3   | 3   | 3   | 3   | 3   | –   | –   | 2   | 1   | –    | 2    |

## List of Experiments / Exercises

| Sl. | Experiment / Exercise                                                                                                            | CO  |
|-----|----------------------------------------------------------------------------------------------------------------------------------|-----|
| 1   | Familiarisation with electrical safety, the multimeter, the oscilloscope, the function generator and the regulated power supply. | CO1 |
| 2   | Verification of Kirchhoff's voltage and current laws.                                                                            | CO1 |
| 3   | Verification of the superposition theorem.                                                                                       | CO1 |
| 4   | Verification of the Thevenin and Norton theorems / the maximum-power-transfer theorem.                                           | CO1 |
| 5   | Study of the series RLC circuit — resonance, the resonant frequency and the quality factor.                                      | CO2 |
| 6   | Measurement of power and power factor in a single-phase circuit and improvement of the power factor by a capacitor.              | CO2 |
| 7   | Open-circuit and short-circuit test on a single-phase transformer / study of a DC machine.                                       | CO4 |
| 8   | Study of the characteristics of the p–n junction diode and the Zener diode.                                                      | CO3 |
| 9   | Study of the half-wave and full-wave rectifier with and without a filter, and the Zener voltage regulator.                       | CO3 |
| 10  | Study of the input and output characteristics of a bipolar junction transistor in the CE configuration.                          | CO3 |
| 11  | Study of the JFET characteristics.                                                                                               | CO3 |
| 12  | Study of the inverting and non-inverting mode of the operational amplifier.                                                      | CO5 |

*A student is to perform not fewer than ten experiments covering all five outcomes, including at least the diode/transistor/JFET characteristics (CO3) and the operational-amplifier experiment (CO5). Every observation is to be recorded against the theory and the measured value compared with the expected.*

## Text and Reference Books

- Laboratory manual, Basic Electrical and Electronics Engineering, MAKAUT (to be issued).
- S.K. Sahdev, Basic Electrical Engineering (with Lab Manual), Khanna Publishing House.
- D. P. Kothari and I. J. Nagrath, Basic Electrical and Electronics Engineering, McGraw-Hill.
- Boylestad and Nashelsky, Electronic Devices and Circuit Theory, Pearson.

## Suggested SWAYAM / NPTEL and Online Resources

- Virtual Labs (Ministry of Education) — Basic Electrical, Electrical Machines and Analog Electronics laboratories.

## Indicative Assessment Map

| Component                               | Coverage                                                                                                                       | Outcomes      |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------|
| Practical CA – I (PCA1, 40)             | First cycle — electrical circuits, AC and machines: conduct 10, observation and data 10, result and interpretation 10, viva 10 | CO1, CO2, CO4 |
| Practical CA – II (PCA2, 40)            | Second cycle — semiconductor devices and the operational amplifier, on the same rubric                                         | CO3, CO5      |
| End-Semester Practical Examination (60) | Record, performance and viva-voce before an external examiner                                                                  | All COs       |

## 26-ES-ME191— Engineering Graphics and Design

|                     |                                 |                      |                            |
|---------------------|---------------------------------|----------------------|----------------------------|
| <b>Course Code</b>  | 26-ES-ME191                     | <b>Category</b>      | Engineering Science Course |
| <b>Course Title</b> | Engineering Graphics and Design | <b>L – T – P</b>     | 0 – 0 – 4                  |
| <b>Credits</b>      | 2                               | <b>Contact Hours</b> | 48                         |
| <b>Semester</b>     | First or Second (by group)      | <b>Assessment</b>    | Practical: PCA 40 + ESE 60 |

### Prerequisites

A two-credit graphics course of about forty-eight hours, rotating with Workshop Practice across the two semesters by group. It assumes only school geometry and begins from the drawing standard. Assumed:

- Plane and solid geometry and coordinate geometry in three dimensions (WBBSE / CBSE / ICSE; CBSE XII Unit IV).
- No prior drawing experience; the course begins from the standard (BIS SP 46) and the instruments.
- The safety and housekeeping induction of the drawing hall / CAD laboratory.

### Course Objectives

- To establish engineering drawing as the graphical language of engineering to the Bureau of Indian Standards code.
- To develop competence in orthographic and isometric projection.
- To introduce sectioning, dimensioning and the drawing of simple machine and civil elements.
- To introduce computer-aided drafting as the modern medium.

### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                                                   | Bloom        |
|-----|-------------------------------------------------------------------------------------------------------------|--------------|
| CO1 | Apply the conventions of engineering drawing and construct the standard geometrical and engineering curves. | L3 – Apply   |
| CO2 | Draw the orthographic projection of a point, line, plane and solid.                                         | L3 – Apply   |
| CO3 | Construct isometric and sectional views and dimension a drawing to the standard.                            | L4 – Analyse |
| CO4 | Produce a drawing using a computer-aided drafting package.                                                  | L3 – Apply   |

### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | 3   | 2   | 3   | 1   | 3   | –   | –   | 1   | 2   | –    | 2    |
| CO2     | 3   | 2   | 3   | 1   | 2   | –   | –   | 1   | 2   | –    | 2    |
| CO3     | 3   | 3   | 3   | 1   | 3   | –   | –   | 1   | 2   | –    | 2    |
| CO4     | 2   | 2   | 3   | 1   | 3   | –   | –   | 2   | 2   | –    | 2    |

## List of Experiments / Exercises

| Sl. | Experiment / Exercise                                                                                               | CO  |
|-----|---------------------------------------------------------------------------------------------------------------------|-----|
| 1   | Introduction to engineering drawing — instruments, the BIS code (SP 46), lettering, lines and dimensioning; scales. | CO1 |
| 2   | Conic sections — the ellipse, parabola and hyperbola; the cycloid and the involute.                                 | CO1 |
| 3   | Orthographic projection — projection of points and of straight lines.                                               | CO2 |
| 4   | Projection of planes.                                                                                               | CO2 |
| 5   | Projection of solids — prism, pyramid, cylinder and cone in simple position.                                        | CO2 |
| 6   | Section of solids and the true shape of the section.                                                                | CO3 |
| 7   | Development of surfaces of simple solids.                                                                           | CO3 |
| 8   | Isometric projection and isometric view of simple solids and their combinations.                                    | CO3 |
| 9   | Conversion of a pictorial view to orthographic views and vice versa.                                                | CO3 |
| 10  | Introduction to a CAD package — the drawing environment, drawing and editing commands.                              | CO4 |
| 11  | Two-dimensional drawing of a simple engineering component in CAD, with dimensioning.                                | CO4 |
| 12  | Introduction to three-dimensional modelling of a simple part in CAD.                                                | CO4 |

*The plates are drawn on the board for the projection modules and in the CAD laboratory for the last three. A student is to complete not fewer than ten plates. The end-semester examination has a board component and a CAD component.*

## Text and Reference Books

- Pradeep Jain, Engineering Graphics & Design, Khanna Publishing House (AICTE Prescribed Textbook)
- N. D. Bhatt, Engineering Drawing, Charotar Publishing.
- K. Venugopal, Engineering Drawing and Graphics, New Age.
- Bureau of Indian Standards, SP 46 — Engineering Drawing Practice.
- Sham Tickoo, AutoCAD / Fusion 360 for Engineers (as adopted), CAD/CIM.

## Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Engineering Drawing and Computer Graphics (IITs), 12 weeks.
- NPTEL — Engineering Graphics, as offered.
- SWAYAM — CAD-based courses, as approved.

## Indicative Assessment Map

| Component                               | Coverage                                                                                   | Outcomes |
|-----------------------------------------|--------------------------------------------------------------------------------------------|----------|
| Practical CA – I (PCA1, 40)             | First cycle — geometrical constructions and orthographic projection: plates and board work | CO1, CO2 |
| Practical CA – II (PCA2, 40)            | Second cycle — isometric, sections, development and CAD                                    | CO3, CO4 |
| End-Semester Practical Examination (60) | Record, performance and viva-voce before an external examiner                              | All COs  |

## 26-ES-ME192 — Workshop Practice

|                     |                            |                      |                            |
|---------------------|----------------------------|----------------------|----------------------------|
| <b>Course Code</b>  | 26-ES-ME192                | <b>Category</b>      | Engineering Science Course |
| <b>Course Title</b> | Workshop Practice          | <b>L – T – P</b>     | 0 – 0 – 4                  |
| <b>Credits</b>      | 2                          | <b>Contact Hours</b> | 48                         |
| <b>Semester</b>     | First or Second (by group) | <b>Assessment</b>    | Practical: PCA 40 + ESE 60 |

### Prerequisites

A two-credit workshop of about forty-eight hours, rotating with Engineering Graphics across the two semesters by group. It assumes no prior experience and begins from the safety induction. Assumed:

- The workshop safety induction, in the first session; no machine is operated without it.
- No prior workshop experience.
- General science of Class X for the properties of the common materials.

### Course Objectives

- To give hands-on experience of the basic manufacturing and fabrication processes.
- To develop safe practice and the correct use of hand tools and machines.
- To connect the process to the material and to the product.
- To introduce measurement and the reading of a job against a drawing.

### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                      | Bloom        |
|-----|--------------------------------------------------------------------------------|--------------|
| CO1 | Work safely in the workshop and select the correct tool and process for a job. | L3 – Apply   |
| CO2 | Make a job to a given drawing in fitting, carpentry and sheet metal.           | L3 – Apply   |
| CO3 | Perform the elementary welding, machining and casting / smithy operations.     | L3 – Apply   |
| CO4 | Measure a job and compare it with the specification on the drawing.            | L4 – Analyse |

### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | 3   | 2   | 2   | 1   | 2   | 1   | 1   | 2   | 1   | –    | 2    |
| CO2     | 3   | 2   | 3   | 1   | 2   | –   | –   | 2   | 1   | –    | 2    |
| CO3     | 3   | 2   | 2   | 1   | 2   | 1   | 1   | 2   | 1   | –    | 2    |
| CO4     | 2   | 3   | 2   | 2   | 2   | –   | –   | 2   | 1   | –    | 2    |

## List of Experiments / Exercises

| Sl. | Experiment / Exercise                                                                          | CO       |
|-----|------------------------------------------------------------------------------------------------|----------|
| 1   | Workshop safety, the layout of the shops, and the common engineering materials and their uses. | CO1      |
| 2   | Fitting — marking, filing, sawing, drilling and tapping to make a job to a drawing.            | CO2, CO4 |
| 3   | Carpentry — planing, marking and making a simple joint (lap or mortise-and-tenon).             | CO2, CO4 |
| 4   | Sheet-metal work — development, cutting, bending and making a simple article.                  | CO2, CO4 |
| 5   | Welding — arc and gas welding; making a simple butt or lap joint.                              | CO3      |
| 6   | Machining — introduction to the lathe; plain and step turning of a job.                        | CO3, CO4 |
| 7   | Foundry / smithy — mould-making, or the forging of a simple shape.                             | CO3      |
| 8   | Introduction to modern manufacturing — a demonstration of 3D printing / CNC.                   | CO1, CO3 |

*A student is to complete a job in each of the fitting, carpentry, sheet-metal and welding sections and at least one of the machining, foundry or smithy sections. Each job is measured against its drawing (CO4).*

## Text and Reference Books

- Veeranna D.K., Workshop / Manufacturing Practices (with Lab Manual), Khanna Publishing House (AICTE Prescribed Textbook).
- A. K. Sarathe, Engineering Workshop Practice, Khanna Publishing House (AICTE Prescribed Textbook).
- Hajra Choudhury, Elements of Workshop Technology, Vol. I, Media Promoters.
- B. S. Raghuvanshi, A Course in Workshop Technology, Dhanpat Rai.
- W. A. J. Chapman, Workshop Technology, CBS.
- Workshop practice manual, MAKAUT (to be issued).

## Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Manufacturing Process Technology / Introduction to Manufacturing, 8 / 12 weeks.
- Virtual Labs (Ministry of Education) — Workshop and Manufacturing laboratories.

## Indicative Assessment Map

| Component                               | Coverage                                                                                | Outcomes      |
|-----------------------------------------|-----------------------------------------------------------------------------------------|---------------|
| Practical CA – I (PCA1, 40)             | First cycle — fitting, carpentry and sheet metal: jobs, finish and dimensional accuracy | CO1, CO2, CO4 |
| Practical CA – II (PCA2, 40)            | Second cycle — welding, machining and foundry / smithy                                  | CO3, CO4      |
| End-Semester Practical Examination (60) | Record, performance and viva-voce before an external examiner                           | All COs       |

## Part VI — Detailed Syllabi: Sessional and Non-Credit Mandatory Courses

The sessional courses (NCC / NSS / Yoga and Sports, and the Skill Enhancement Course) and the two mandatory non-credit courses (the Indian Knowledge System and the Indian Constitution) follow.

### A. What a sessional course is, and which courses this document covers

A sessional course is assessed wholly by internal, continuous evaluation across the semester, against published rubrics, rather than by a single end-of-semester written theory paper. In the first-year structure the courses below are assessed on a sessional or continuous basis, and are covered here. Marks are out of 100 unless the course is non-credit, in which case it is graded Satisfactory or Unsatisfactory.

| Code       | Course                                    | Credits / L-T-P    | Assessment basis                                                                                        |
|------------|-------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------|
| 26-VAC-181 | NCC / NSS / Yoga and Sports               | 2 (0-0-4)          | Pure sessional — 100 marks, wholly internal continuous evaluation                                       |
| 26-SEC-181 | Skill Enhancement Course — MOOCs / Skills | 1 (0-0-2)          | Sessional / MOOC — 100 marks; MOOC score + applied task + reflection, or institutional skill assessment |
| 26-MC-182  | Indian Knowledge System                   | Non-credit (2-0-0) | Sessional, non-credit — graded Satisfactory / Unsatisfactory                                            |
| 26-MC-181  | Indian Constitution                       | Non-credit (2-0-0) | Sessional, non-credit — graded Satisfactory / Unsatisfactory                                            |

### B. General principles governing every sessional assessment

| Principle                   | What it means                                                                                                                                                                                                                               |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Continuous, not terminal    | Assessment is distributed across the whole semester in at least two cycles; no sessional course is settled by a single end-point exercise.                                                                                                  |
| Rubric published in advance | Every qualitative component is marked against a written rubric (the Rubric Bank at the end of this document) given to the student before the task is set.                                                                                   |
| Assessor and moderation     | The subject teacher or stream in-charge assesses; a sessional committee of the department moderates. An external examiner is associated with the final evaluation of every credit-bearing sessional course, but not the non-credit courses. |
| Minimum conditions          | Attendance of not less than 75% is a necessary but not sufficient condition of a pass; the student must also secure at least 40% of the sessional marks, or, for a non-credit course, a Satisfactory grade.                                 |
| Outcome attainment          | Course-Outcome attainment is computed as in Part I of the syllabus — 80% from the direct marks mapped to each outcome and 20% from an outcome-based student feedback instrument.                                                            |
| Records for NBA / NAAC      | Attendance registers, rubric sheets, log books, artefacts, certificates and the assessment sheet are retained for not fewer than three assessment years, as the NBA continuous-improvement criterion requires.                              |
| Uniformity across colleges  | The scheme, the components and the rubrics are uniform across every affiliated college; a college may not vary the marks distribution.                                                                                                      |

## 26-VAC-181 — NCC / NSS / Yoga and Sports

|                     |                             |                      |                                                        |
|---------------------|-----------------------------|----------------------|--------------------------------------------------------|
| <b>Course Code</b>  | 26-VAC-181                  | <b>Category</b>      | Value-Added Course                                     |
| <b>Course Title</b> | NCC / NSS / Yoga and Sports | <b>L – T – P</b>     | 0 – 0 – 4                                              |
| <b>Credits</b>      | 2                           | <b>Contact Hours</b> | 52                                                     |
| <b>Semester</b>     | First                       | <b>Assessment</b>    | Sessional: wholly internal continuous evaluation (100) |

### Prerequisites

A two-credit activity-based value-added course. The student chooses one stream — the National Cadet Corps, the National Service Scheme, or Yoga and Sports — and is assessed on participation and a record of activity. Assumed: none beyond the medical fitness the activity requires.

- No academic prerequisite.
- A declaration of medical fitness for the physical activity of the chosen stream.
- Enrolment in one stream at the start of the semester.

### Course Objectives

- To develop physical fitness, discipline and the habit of regular activity.
- To develop service, citizenship and community engagement.
- To develop teamwork, leadership and resilience.
- To contribute to the holistic development the National Education Policy seeks.

### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                                  | Bloom        |
|-----|--------------------------------------------------------------------------------------------|--------------|
| CO1 | Participate regularly and safely in the physical or service activity of the chosen stream. | L3 – Apply   |
| CO2 | Demonstrate discipline, teamwork and responsibility in the activity.                       | L3 – Apply   |
| CO3 | Reflect on the value of service, fitness or citizenship through a record of activity.      | L4 – Analyse |

### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | 1   | –   | –   | –   | –   | 2   | 2   | 3   | 2   | 1    | 2    |
| CO2     | –   | –   | –   | –   | –   | 2   | 2   | 3   | 2   | 1    | 2    |
| CO3     | 1   | –   | –   | –   | –   | 3   | 2   | 2   | 2   | 1    | 3    |

## List of Experiments / Exercises

| Sl. | Experiment / Exercise                                                                                                    | CO       |
|-----|--------------------------------------------------------------------------------------------------------------------------|----------|
| 1   | NCC stream — drill, discipline, national-security awareness, and camp / social-service activity, per the NCC syllabus.   | CO1, CO2 |
| 2   | NSS stream — orientation, regular activity, and a special camp of community service, per the NSS framework.              | CO1, CO3 |
| 3   | Yoga and Sports stream — yoga practice (āsana, prāṇāyāma) and its documented basis, and participation in a chosen sport. | CO1, CO2 |
| 4   | A record / log of activity across the semester, and a short reflective report.                                           | CO3      |

*Each student follows one stream for the whole semester. Assessment is wholly internal and continuous — attendance and regularity, participation and conduct, camp or event participation, and the activity record and reflective report. A minimum attendance and participation is a necessary condition of a pass.*

### Text and Reference Books

- NCC Cadet Handbook, Directorate General of NCC (for the NCC stream).
- National Service Scheme Manual, Ministry of Youth Affairs and Sports (for the NSS stream).
- A textbook of Yoga and the Common Yoga Protocol, Ministry of AYUSH (for the Yoga stream).

### Suggested SWAYAM / NPTEL and Online Resources

- SWAYAM / NPTEL — Science of Yoga / Yoga and Wellness, as offered.
- Fit India and MY Bharat resources, for activity and reporting.

## 26-VAC-181 — NCC / NSS / Yoga and Sports (2 credits, 0-0-4, Semester I)

A pure sessional course of 100 marks, wholly internal and continuous. The student follows one stream for the whole semester; the components are the same across the three streams, the stream-specific activity differing only in kind. An external examiner is associated with the final evaluation.

### Course Outcomes assessed

| CO  | Statement                                                                                  |
|-----|--------------------------------------------------------------------------------------------|
| CO1 | Participate regularly and safely in the physical or service activity of the chosen stream. |
| CO2 | Demonstrate discipline, teamwork and responsibility in the activity.                       |
| CO3 | Reflect on the value of service, fitness or citizenship through a record of activity.      |

### Assessment components and marks

| Sl. | Component                          | Marks | Weight | How it is assessed (rubric / evidence)                                                                                                                              |
|-----|------------------------------------|-------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Attendance and regularity          | 20    | 20%    | From the attendance register across the semester; 75% attendance is the qualifying floor, marks scaled above it. (CO1)                                              |
| 2   | Participation and conduct          | 20    | 20%    | Continuous observation of engagement, discipline and responsibility, on the four-band rubric. (CO2)                                                                 |
| 3   | Activity / skill performance       | 30    | 30%    | Stream-specific: NCC — drill and turnout; NSS — service-activity performance; Yoga & Sports — āsana / prāṇāyāma practice and sport skill, on the rubric. (CO1, CO2) |
| 4   | Camp / event / field participation | 15    | 15%    | NCC camp / NSS special camp / a sports meet or yoga demonstration, evidenced by a certificate or supervisor report. (CO1)                                           |
| 5   | Activity log and reflective report | 15    | 15%    | A maintained log of activity and a short reflective report, on the reflection rubric. (CO3)                                                                         |
|     | Total                              | 100   | 100%   |                                                                                                                                                                     |

### Minimum requirements to pass

- Attendance not less than 75% (a necessary condition).
- At least 40% in the aggregate (40 of 100).
- Participation in at least one camp, event or field activity of the chosen stream.
- Submission of the activity log and reflective report.

### Assessor, moderation and records

| Assessed and moderated by                                                                                                                                                                                                                             | Evidence retained (NBA / NAAC)                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The NCC Associate NCC Officer, the NSS Programme Officer, or the Physical Education / Yoga instructor as the stream in-charge, internally; moderated by the departmental sessional committee; external examiner associated with the final evaluation. | Attendance register; participation and conduct observation sheet; activity / skill rubric sheet; camp or event certificate or supervisor report; the activity log; the reflective report; and the consolidated assessment sheet. |

## 26-SEC-181 (0-0-2, 1)— Skill Enhancement Course — MOOCs / Skills

|                     |                                           |                      |                                                 |
|---------------------|-------------------------------------------|----------------------|-------------------------------------------------|
| <b>Course Code</b>  | 26-SEC-181 (0-0-2, 1)                     | <b>Category</b>      | Skill Enhancement Course                        |
| <b>Course Title</b> | Skill Enhancement Course — MOOCs / Skills | <b>L – T – P</b>     | See note                                        |
| <b>Credits</b>      | 1                                         | <b>Contact Hours</b> | As per the credit of the chosen course          |
| <b>Semester</b>     | First and Second                          | <b>Assessment</b>    | Sessional: as per the mode of the chosen course |

### Prerequisites

A skill course taken from a specified, published basket — through SWAYAM Plus, Future Skills Prime, an approved MOOC, or a practical / hands-on / soft-skill course offered by the institution or University. SEC-181 in Semester I carries one credit; SEC-282 in Semester II carries two credits. The Department publishes the basket, with the outcomes, mode, effort, assessment and fallback of each entry, before registration each session; the student chooses from the basket, not from the whole of a platform.

- The prerequisite of the specific course chosen, as stated in the published basket.
- Registration on the platform, and mentor allocation by the Department.
- For a credit-bearing MOOC, the proctored examination the platform requires.

### Course Objectives

- To bring an employable practical skill to the student from the first year.
- To place at least part of the first year on a national digital-learning platform, on the University's early-mover credit-transfer policy.
- To let the student begin to shape the degree towards an interest or a specialisation.
- To do so under a specified and quality-assured basket rather than an open choice.

### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                         | Bloom        |
|-----|-----------------------------------------------------------------------------------|--------------|
| CO1 | Complete an approved skill or MOOC course and demonstrate the skill it certifies. | L3 – Apply   |
| CO2 | Apply the skill to a small task, problem or artefact.                             | L3 – Apply   |
| CO3 | Reflect on the learning and relate it to the programme and to employability.      | L4 – Analyse |

### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | 2   | 1   | 1   | 1   | 3   | –   | –   | 1   | 1   | –    | 3    |
| CO2     | 2   | 2   | 2   | 1   | 3   | –   | –   | 1   | 1   | –    | 3    |
| CO3     | 1   | 1   | 1   | 1   | 2   | 1   | 1   | 1   | 1   | –    | 3    |

## List of Experiments / Exercises

| Sl. | Experiment / Exercise                                                                                                                                                                                                       | CO  |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1   | An indicative basket — digital productivity and office tools; an introduction to Python; computer-aided drafting; an introduction to the Internet of Things; a communication, entrepreneurship or financial-literacy skill. | CO1 |
| 2   | Completion of the chosen course with its certificate or the institutional assessment.                                                                                                                                       | CO1 |
| 3   | A small applied task, artefact or portfolio demonstrating the skill.                                                                                                                                                        | CO2 |
| 4   | A short reflective note relating the skill to the programme.                                                                                                                                                                | CO3 |

*The credit is awarded on completion of the chosen course and the associated applied task, verified by the mentor. For a SWAYAM course the platform's proctored score is used; for an institutional course the internal assessment is used. The specified basket, with outcomes and a named fallback against each entry, is a matter for the Board (Part X) and is to be published before registration.*

## Text and Reference Books

- As prescribed by the specific course chosen from the published basket.

## Suggested SWAYAM / NPTEL and Online Resources

- SWAYAM Plus and SWAYAM — the approved credit-bearing courses of the published basket.
- Future Skills Prime (NASSCOM – MeitY) — approved digital-skill courses.
- NPTEL — short (4 / 8-week) skill courses aligned to the basket, as approved.

## 26-SEC-181 (1 cr, 0-0-2, Sem I) — Skill Enhancement Course (MOOCs / Skills)

A sessional course taken from the specified, published basket, in one of two modes — an approved SWAYAM / NPTEL MOOC, or an institutional skill course. It carries 100 marks in either mode. SEC-181 is the shorter, one-credit course and SEC-282 the fuller, two-credit course. The department mentor verifies completion; for a MOOC the platform's proctored score is authoritative for its part.

### Course Outcomes assessed

| CO  | Statement                                                                         |
|-----|-----------------------------------------------------------------------------------|
| CO1 | Complete an approved skill or MOOC course and demonstrate the skill it certifies. |
| CO2 | Apply the skill to a small task, problem or artefact.                             |
| CO3 | Reflect on the learning and relate it to the programme and to employability.      |

### Assessment components and marks

| Sl. | Component                             | Marks | Weight | How it is assessed (rubric / evidence)                                                                                                                           |
|-----|---------------------------------------|-------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Course completion and proctored score | 60    | 60%    | MOOC mode: the platform's proctored final score, scaled to 60. Institutional mode: a continuous skill assessment (40) and a skill-demonstration test (20). (CO1) |
| 2   | Applied task / artefact               | 25    | 25%    | A task, artefact or mini-project applying the skill, verified by the mentor against a rubric. (CO2)                                                              |
| 3   | Reflective note                       | 15    | 15%    | A short note relating the skill to the programme and to employability, on the reflection rubric. (CO3)                                                           |
|     | Total                                 | 100   | 100%   |                                                                                                                                                                  |

### The two delivery modes at a glance

| Mode                                        | Component 1 (60)                                 | Comp. 2 (25) | Comp. 3 (15) | Best suited to                                            |
|---------------------------------------------|--------------------------------------------------|--------------|--------------|-----------------------------------------------------------|
| MOOC (SWAYAM / NPTEL / Future Skills Prime) | Proctored platform score, scaled to 60           | Applied task | Reflection   | A college without the faculty to teach the skill in house |
| Institutional skill course                  | Continuous assessment 40 + demonstration test 20 | Applied task | Reflection   | A skill the college can deliver and assess directly       |

### Minimum requirements to pass

- MOOC mode: the platform completion certificate with a passing proctored score. Institutional mode: at least 40% in the skill assessment.
- Mentor verification of the applied task is mandatory in both modes.
- At least 40% in the aggregate (40 of 100).
- The chosen course must be from the basket published by the department before registration.

### Assessor, moderation and records

| Assessed and moderated by                                                                                                                                                                                                  | Evidence retained (NBA / NAAC)                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The department-allocated mentor, internally; moderated by the sessional committee. For a MOOC the SWAYAM / NPTEL proctored score is used directly for component 1. External examiner associated with the final evaluation. | The published basket for the session; the platform certificate and score report (MOOC mode) or the skill-assessment sheet (institutional mode); the applied task / artefact; the reflective note; the mentor verification; and the consolidated assessment sheet. |

## 26-MC-181— Indian Constitution

|                     |                            |                      |                                           |
|---------------------|----------------------------|----------------------|-------------------------------------------|
| <b>Course Code</b>  | 26-MC-181                  | <b>Category</b>      | Mandatory Course (Non-Credit)             |
| <b>Course Title</b> | Indian Constitution        | <b>L – T – P</b>     | 2 – 0 – 0                                 |
| <b>Credits</b>      | Nil                        | <b>Contact Hours</b> | 24                                        |
| <b>Semester</b>     | First or Second (by group) | <b>Assessment</b>    | Non-credit; Satisfactory / Unsatisfactory |

### Prerequisites

A mandatory non-credit course, paired with the Indian Knowledge System across the two semesters and rotated by group. It is self-contained and assumes only the civics of Class X.

- No prerequisite beyond the civics and social science of Class X.
- No prior study of law is assumed.
- The course is descriptive and civic in purpose, not a course in legal practice.

### Course Objectives

- To acquaint the student with the making, structure and principles of the Constitution of India.
- To establish the fundamental rights and duties and the directive principles.
- To outline the structure of the Union, the States and the local bodies.
- To develop the constitutional and civic awareness the graduate engineer needs.

### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                                | Bloom            |
|-----|------------------------------------------------------------------------------------------|------------------|
| CO1 | Describe the making, the Preamble, and the basic structure of the Constitution of India. | I.2 – Understand |
| CO2 | Explain the fundamental rights and duties and the directive principles of State policy.  | I.2 – Understand |
| CO3 | Outline the structure and functioning of the Union, the State and the local government.  | I.2 – Understand |

### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | –   | –   | –   | –   | –   | 3   | 3   | 1   | 1   | –    | 1    |
| CO2     | –   | –   | –   | –   | –   | 3   | 3   | 1   | 1   | –    | 1    |
| CO3     | 1   | –   | –   | –   | –   | 3   | 2   | 1   | 1   | –    | 2    |

## Course Plan — Module-wise Syllabus

| Module | Title                                               | Contents                                                                                                                                                                                                                                      | Hours |
|--------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| I      | The Making and Framework of the Constitution        | The historical background and the Constituent Assembly; the Preamble and its ideals; the salient features of the Constitution; the idea of the basic structure.                                                                               | 6     |
| II     | Fundamental Rights, Duties and Directive Principles | The fundamental rights and the writs; the fundamental duties; the directive principles of State policy and their relation to the rights.                                                                                                      | 7     |
| III    | The Union and the States                            | The Union executive, the Parliament and the judiciary; the State executive and legislature; the relation between the Union and the States in outline; the amendment of the Constitution.                                                      | 6     |
| IV     | Local Government and Constitutional Bodies          | The Panchayati Raj and the municipalities; the principal constitutional bodies — the Election Commission, the Union and State Public Service Commissions, the Comptroller and Auditor-General — in outline; the citizen and the Constitution. | 5     |

**Non-credit character.** The course carries no credit and no marks contribute to the Semester Grade Point Average. It is a mandatory non-credit course, paired with the Indian Knowledge System across the two semesters and rotated by group, and a student must be graded Satisfactory to be eligible for the award of the degree. The grade is awarded by the college on attendance of not less than seventy-five per cent, one assignment, and one class test of twenty-five marks in which at least forty per cent is to be obtained. A student graded Unsatisfactory is re-examined in a following semester. The course gives effect to the AICTE mandatory-course requirement on the Constitution of India.

### Text and Reference Books

- Durga Das Basu, Introduction to the Constitution of India, LexisNexis.
- M. V. Pylee, India's Constitution, S. Chand.
- Subhash Kashyap, Our Constitution, National Book Trust.
- AICTE material on the Indian Constitution mandatory course.

### Suggested SWAYAM / NPTEL and Online Resources

- SWAYAM / NPTEL — Introduction to the Constitution of India, as offered.
- Resources of the Ministry of Law and Justice and the Election Commission of India.

Mandatory non-credit courses, paired across the two semesters and rotated by group. No marks contribute to the Semester Grade Point Average; each is graded Satisfactory or Unsatisfactory. A Satisfactory grade in both is a condition of the award of the degree.

### Grading components (both courses)

| Component          | What it consists of                                                                                                                                                                                  | Requirement                               | Course outcome                 |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------|
| Attendance         | Attendance across the semester                                                                                                                                                                       | Not less than 75% (a necessary condition) | CO1–CO3                        |
| Assignment / essay | IKS: a sourced essay of about a thousand words on a documented Indian contribution relevant to the branch, and a site or museum visit report.<br>Constitution: one assignment on a prescribed theme. | Both submitted and judged acceptable      | IKS CO2, CO3 / Const. CO2, CO3 |
| Class test         | One class test of 25 marks over the course                                                                                                                                                           | At least 40% (10 of 25)                   | CO1, CO2                       |

### The Satisfactory / Unsatisfactory decision

- Satisfactory (S): attendance not less than 75%, both the assignment / essay (and, for IKS, the visit report) submitted and acceptable, and at least 50% in the class test.
- Unsatisfactory (US): any one of the above not met.
- A student graded Unsatisfactory is re-examined or re-assessed in a following semester; the grade does not enter the SGPA either way.
- A Satisfactory grade in both mandatory courses, across the two semesters, is a necessary condition of the award of the degree.

*Assessed by the subject teacher and recorded by the college; no external examiner is required for a non-credit course. Records retained: the attendance register, the essay or assignment and visit report, the class-test scripts, and the Satisfactory / Unsatisfactory award sheet. Although non-credit, the Indian Knowledge System course remains the foundational IKS course required by the UGC Guidelines of 2023.*

## 26-MC-182 — Indian Knowledge System

|                     |                            |                      |                                           |
|---------------------|----------------------------|----------------------|-------------------------------------------|
| <b>Course Code</b>  | 26-MC-182                  | <b>Category</b>      | Mandatory Course (Non-Credit)             |
| <b>Course Title</b> | Indian Knowledge System    | <b>L – T – P</b>     | 2 – 0 – 0                                 |
| <b>Credits</b>      | Nil                        | <b>Contact Hours</b> | 24                                        |
| <b>Semester</b>     | First or Second (by group) | <b>Assessment</b>    | Non-credit; Satisfactory / Unsatisfactory |

### Prerequisites

A mandatory non-credit course, paired with the Indian Constitution across the two semesters and rotated by group. It is self-contained and assumes no prior study and no knowledge of Sanskrit.

- No prerequisite; the course is introductory.
- No knowledge of Sanskrit is assumed; all primary material is read in translation.
- A general Class X familiarity with Indian history and geography is sufficient.

### Course Objectives

- To introduce the classification, sources and vocabulary of the Indian knowledge traditions.
- To survey the Indian contribution to mathematics, astronomy, the sciences of materials and construction, and health and environment.
- To introduce the methods by which such knowledge was recorded and transmitted.
- To prepare the ground for the discipline-embedded Indian Knowledge System modules of the later years.

### Course Outcomes

On completion of the course the student will be able to:

| CO  | Statement                                                                                                             | Bloom           |
|-----|-----------------------------------------------------------------------------------------------------------------------|-----------------|
| CO1 | Describe the classification, principal sources and vocabulary of the Indian knowledge traditions.                     | L2 – Understand |
| CO2 | Give examples of the Indian contribution to mathematics, science, materials and construction relevant to engineering. | L2 – Understand |
| CO3 | Relate an element of the Indian knowledge tradition to the student's own branch of engineering.                       | L3 – Apply      |

### CO–PO Articulation Matrix

*Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).*

| CO \ PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1     | 1   | 1   | –   | –   | –   | 2   | 2   | 1   | 1   | –    | 2    |
| CO2     | 2   | 1   | 1   | –   | –   | 2   | 2   | 1   | 1   | –    | 2    |
| CO3     | 2   | 2   | 2   | –   | –   | 3   | 2   | 1   | 1   | –    | 3    |

### Course Plan — Module-wise Syllabus

| Module | Title                                       | Contents                                                                                                                                                                                                                                                                                      | Hours |
|--------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| I      | Introduction to the Indian Knowledge System | The classification of the Indian knowledge traditions; the vidyāsthāna scheme; the primary sources and the vocabulary; the sūtra–vṛtti–bhāṣya form of recording knowledge; the oral and the manuscript tradition and their preservation.                                                      | 5     |
| II     | Mathematics and Astronomy                   | The Indian number system and the place-value notation; śulba-sūtra geometry; the algorithmic tradition — Āryabhaṭa, Brahmagupta, Bhāskara; Piṅgala's combinatorics and binary enumeration; Indian astronomy and the yantras of the observatories, in outline.                                 | 7     |
| III    | Materials, Construction and Technology      | Metallurgy — wootz (crucible) steel and the corrosion-resistant iron; construction — the Śilpa- and Vāstu-śāstra, temple and civic architecture, lime mortars and standardised brick; water — stepwells, tanks and traditional harvesting; the craft traditions and the śreṇī (guild) system. | 7     |

| Module | Title                                              | Contents                                                                                                                                                                                                                                                            | Hours |
|--------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| IV     | Health, Environment and the Engineering Connection | Āyurveda and the traditions of health in outline; the environmental and sustainability logic of the Indian traditions; the connection of the Indian knowledge tradition to the student's own branch, previewing the discipline-embedded modules of the later years. | 5     |

**Non-credit character.** The course carries no credit and no marks contribute to the Semester Grade Point Average. It is a mandatory non-credit course, paired with the Indian Constitution across the two semesters and rotated by group, and a student must be graded Satisfactory to be eligible for the award of the degree. The grade is awarded by the college on attendance of not less than seventy-five per cent, one sourced short essay of about a thousand words on a documented Indian contribution relevant to the student's branch, one site or museum visit report, and one class test of twenty-five marks in which at least forty per cent is to be obtained. A student graded Unsatisfactory is re-examined in a following semester. Although non-credit, it is the foundational Indian Knowledge Systems course required by the UGC Guidelines of 2023 and the anchor of the discipline-embedded IKS modules of the later years.

### Text and Reference Books

- M. C. Bora, Foundations of Bhartiya Knowledge Systems, Khanna Publishing House.
- B. Mahadevan, Vinayak Rajat Bhat and Nagendra Pavana R. N., Introduction to Indian Knowledge System: Concepts and Applications, PHI.
- Gautam R. Desiraju, Indian Knowledge System, Khanna Publishing House.
- Kapil Kapoor and Avadhesh Kumar Singh (eds.), Indian Knowledge Systems, D. K. Printworld.
- Subhash Vats, Religion & Dharma, Khanna Book Publishing Co.
- Pride of India: A Glimpse into India's Scientific Heritage, Samskrita Bharati.
- AICTE / Ministry of Education material on the Indian Knowledge System.

### Suggested SWAYAM / NPTEL and Online Resources

- SWAYAM / NPTEL — Introduction to Indian Knowledge System / Indian Culture and Heritage, as offered.
- Resources of the IKS Division, Ministry of Education (iksindia.org).

Mandatory non-credit courses, paired across the two semesters and rotated by group. No marks contribute to the Semester Grade Point Average; each is graded Satisfactory or Unsatisfactory. A Satisfactory grade in both is a condition of the award of the degree.

### Grading components (both courses)

| Component          | What it consists of                                                                                                                                                                                  | Requirement                               | Course outcome                 |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------|
| Attendance         | Attendance across the semester                                                                                                                                                                       | Not less than 75% (a necessary condition) | CO1–CO3                        |
| Assignment / essay | IKS: a sourced essay of about a thousand words on a documented Indian contribution relevant to the branch, and a site or museum visit report.<br>Constitution: one assignment on a prescribed theme. | Both submitted and judged acceptable      | IKS CO2, CO3 / Const. CO2, CO3 |
| Class test         | One class test of 25 marks over the course                                                                                                                                                           | At least 40% (10 of 25)                   | CO1, CO2                       |

### The Satisfactory / Unsatisfactory decision

- Satisfactory (S): attendance not less than 75%, both the assignment / essay (and, for IKS, the visit report) submitted and acceptable, and at least 50% in the class test.
- Unsatisfactory (US): any one of the above not met.
- A student graded Unsatisfactory is re-examined or re-assessed in a following semester; the grade does not enter the SGPA either way.
- A Satisfactory grade in both mandatory courses, across the two semesters, is a necessary condition of the award of the degree.

*Assessed by the subject teacher and recorded by the college; no external examiner is required for a non-credit course. Records retained: the attendance register, the essay or assignment and visit report, the class-test scripts, and the Satisfactory / Unsatisfactory award sheet. Although non-credit, the Indian Knowledge System course remains the foundational IKS course required by the UGC Guidelines of 2023.*

## Rubric Bank — the four-band scale for the qualitative components

Every qualitative component named in the schemes above is marked on this four-band scale. The band gives the fraction of the component's marks, and is published to the student before the task is set.

| Sr.No | Criterion                              | Excellent (90–100%)                             | Good (70–89%)            | Satisfactory (50–69%)           | Needs Improvement (<50%)       |
|-------|----------------------------------------|-------------------------------------------------|--------------------------|---------------------------------|--------------------------------|
| 1     | Participation and regularity           | Present and fully engaged throughout; initiates | Regular and engaged      | Present, limited engagement     | Irregular or passive           |
| 2     | Discipline, conduct and responsibility | Exemplary; a model to peers                     | Consistently responsible | Generally acceptable            | Repeated lapses                |
| 3     | Skill / activity performance           | Skilful, accurate, safe; near-independent       | Competent and safe       | Basic competence with guidance  | Below the expected standard    |
| 4     | Teamwork and interpersonal             | Leads and enables the team                      | Collaborates effectively | Works in the team when directed | Works in isolation or disrupts |
| 5     | Communication (oral / GD / interview)  | Clear, fluent, confident, well-structured       | Clear and organised      | Understandable, some hesitation | Unclear or disorganised        |
| 6     | Task / artefact quality                | Complete, correct, well-presented, original     | Complete and correct     | Mostly complete, minor errors   | Incomplete or incorrect        |
| 7     | Reflection quality                     | Insightful; links learning to the programme     | Thoughtful and relevant  | Descriptive, some reflection    | Superficial or absent          |
| 8     | Record / log / portfolio               | Thorough, orderly, reproducible                 | Complete and orderly     | Adequate                        | Sparse or disorganised         |

### Applying the bands

- The assessor selects one band per criterion and awards the mid-point of the band unless a specific justification is recorded; the component mark is the sum, or weighted mean, across its criteria.
- For the NCC / NSS / Yoga participation, conduct and skill components, criteria 1–4 apply; for the SEC and the AEC courses, criteria 5–7; the record / log criterion applies wherever a log or portfolio is required.
- A copy of the completed rubric sheet is retained with the assessment record for audit.

## Part VII — Sources and Benchmarking

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### VII.1 Policy and regulatory instruments

- Ministry of Education, Government of India. National Education Policy 2020.
- University Grants Commission. Curriculum and Credit Framework for Undergraduate Programmes, 2022.
- University Grants Commission. Guidelines for Incorporating Indian Knowledge System in Higher Education Curricula, 2023; and the draft Guidelines and Curriculum Framework on Mission LiFE at the Undergraduate Level.
- All India Council for Technical Education. Model Curriculum for Undergraduate Degree Courses in Engineering and Technology — the first-year common syllabus, Universal Human Values and the mandatory course list.
- National Board of Accreditation. GAPC v4.0 accreditation framework (Washington Accord aligned) — the eleven Programme Outcomes of Part I and the attainment framework.
- National Assessment and Accreditation Council. Revised Assessment and Accreditation Framework — Criteria 1 and 2.
- Bureau of Indian Standards, SP 46 — Engineering Drawing Practice, for the Graphics course.
- Direction of the Hon'ble Supreme Court of India on compulsory environmental education, and the consequent UGC syllabus.

### VII.2 University documents

- MAKAUT, WB. Finalised NEP 2020 Compliant Curriculum Structure for the First Year (B.Tech. First Year, MAKAUT 2026), effective A.Y. 2026–27 — reproduced in Part V.
- MAKAUT, WB. Suggested modifications to the Basic Electrical and Electronics Engineering theory and laboratory syllabus — incorporated in ES-EE101 and ES-EE191.
- MAKAUT, WB. Detailed syllabus of ES-CS201 Programming for Problem Solving (with the Introduction to Artificial Intelligence module) — incorporated in ES-CS201 and ES-CS291.

### VII.3 School syllabi consulted for the prerequisite mapping

- West Bengal Council of Higher Secondary Education — Mathematics, Physics, Chemistry, Biology, Computer Application and English, Classes XI and XII.
- Central Board of Secondary Education — the corresponding subjects, Classes XI and XII.
- Council for the Indian School Certificate Examinations — the ISC syllabuses for the corresponding subjects.
- West Bengal Board of Secondary Education — Class X Science and Mathematics.

### VII.4 Institutional syllabi consulted

The first-year syllabi of the following were examined for the module sequence, depth and treatment of the common first year. No syllabus has been adopted; the responsibility for the content is the University's.

- Indian Institute of Technology Kharagpur and Madras — first-year core courses, the Physics and Chemistry laboratories, and Problem Solving through Programming.
- Jadavpur University, Faculty of Engineering and Technology — first-year Mathematics, Physics, Chemistry and Basic Electrical and Electronics.
- National Institute of Technology Durgapur and IIST Shibpur — first-year common curriculum, workshop and graphics.
- Anna University Regulation 2023 and Visvesvaraya Technological University 2022 scheme — NEP-aligned first-year structures and the Physics/Chemistry group rotation.
- AICTE model first-year cycle-rotation scheme, and the AICTE model syllabus for the Introduction to Artificial Intelligence content.